

PMath 365: Differential Geometry

Fixing notation

Most of the course we'll be working in $\mathbb{R}^n$, with points typically denoted by $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ or $y = (y_1, \dots, y_n) \in \mathbb{R}^n$, always using the Euclidean norm $\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$ and Euclidean distance $d(x, y) = \|y - x\|$. The open ball for radius $r > 0$ will be denoted by $B_r(x) = \{y \in \mathbb{R}^n \mid d(x, y) < r\}$, and the closed ball for radius $r > 0$ will be denoted by $\overline{B}_r(x) = \{y \in \mathbb{R}^n \mid \|y - x\| \leq r\}$.

Chapter 1: Curves

Definition

A subset of $\mathbb{R}^n$ that can be parameterized in one variable $t \in (a, b) \subseteq \mathbb{R}$ is called a curve.

Example

Graphs of (continuous) functions $f : (a, b) \subseteq \mathbb{R} \rightarrow \mathbb{R}^n$ mapping $x \mapsto f(x)$ give a graph $G = \{(x, y) \in \mathbb{R}^{n+1} : y = f(x)\} \subseteq \mathbb{R}^{n+1}$. Then G can be parameterized by $\gamma : (a, b) \rightarrow \mathbb{R}^n$ acting by $\gamma(t) = (t, f(t))$.

For example, for a curve, we can take the graph of $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, for which G is a parabola. We note that f , and thus γ is differentiable everywhere in this example.

As another example, consider the graph of the function $f(x) = |x|$. Note that f is not differentiable at 0.

Example

The level set of a function $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, U open, for $k \in \text{im}(f)$, then $f^{-1}(k)$ is a curve if f 's total derivative has full rank at all points of $f^{-1}(k)$ (by the implicit function theorem). As an example, this works for the function $f(x, y) = x^2 + y^2$ for all $k > 0$ (this curve is the circle of radius $\sqrt{k}$), which can be parameterized by many of the known parameterizations of the circle.

Example

In $\mathbb{R}^3$, some examples of curves include the 'nice' intersections of two surfaces, like the intersection of the paraboloid $z = x^2 + y^2$ and the horizontal plane $z = 2$ gives a circle at height 2, parameterizable by $\gamma(t) = (\sqrt{2} \cos t, \sqrt{2} \sin t, 2), t \in \mathbb{R}$.

We can get another curve, called the twisted cubic, by taking the intersection of $y = x^2$ and $z = x^3$, which is parameterized by $\gamma(t) = (t, t^2, t^3), t \in \mathbb{R}$.

Example

Another important example is the helix, parameterized $\gamma(t) = (a \cos t, a \sin t, bt)$ for fixed $a > 0, b \in \mathbb{R}$. Every point of this curve lies on the cylinder $x^2 + y^2 = a^2$, and twists up or down (depending on the sign of b) in a circular fashion.

We will consider in the course curves that are parameterized by $\gamma : (a, b) \subseteq \mathbb{R} \rightarrow \mathbb{R}^n$ where its component functions are class $\mathcal{C}^r$ for some $r \in \{1, \dots, \infty\}$. Such a curve is called a $\mathcal{C}^r$ curve.

Definition

A parameterized curve $\gamma : (a, b) \rightarrow \mathbb{R}^n$ is called regular if $\gamma'(t) := (\gamma'_1(t), \dots, \gamma'_n(t)) \neq (0, \dots, 0)$ for all $t \in (a, b)$.

Definition

For a parameterized curve $\gamma : (a, b) \rightarrow \mathbb{R}^n$, we call $\gamma'(t)$ the velocity of γ at $\gamma(t)$ and $\|\gamma'(t)\|$ the speed of γ at $\gamma(t)$. A regular curve is one where the speed does not vanish at any point. A parameterized curve is called unit speed if $\|\gamma'(t)\| = 1$ for all $t \in (a, b)$. Note that all unit speed curves are always regular.

Example

The circle $x^2 + y^2 = a^2$ parameterized by $\gamma(t) = (a \cos(ct), a \sin(ct))$ for fixed $a > 0, c \neq 0 \in \mathbb{R}$ is a regular $\mathcal{C}^\infty$ parameterization, since we can check $\|\gamma'(t)\| = a|c| \neq 0$ for all t , and γ is unit-speed if and only if $c = \pm \frac{1}{a}$. Thus we can give a unit-speed parameterization of the circle: $\gamma(t) = (a \cos(t/a), a \sin(t/a))$.

Example

Recall the helix parameterized by $\gamma(t) = (a \cos t, a \sin t, bt)$ for $t \in \mathbb{R}$, where we see $\gamma'(t) = (-a \sin t, a \cos t, b)$ for $a > 0, b \neq 0$. We thus see it is a regular parameterization because b never vanishes. Moreover, $\|\gamma'(t)\| = \sqrt{a^2 + b^2} > 0$, implying γ is unit-speed if and only if $\sqrt{a^2 + b^2} = 1$.

Example

Recall the twisted cubic parameterized by $\gamma(t) = (t, t^2, t^3)$. Then $\gamma'(t) = (1, 2t, 3t^2) \neq 0$, so it is regular, but $\|\gamma'(t)\| = \sqrt{1 + 4t^2 + 9t^4} \geq 1$ for all $t \in \mathbb{R}$, but only has unit-speed at $t = 0$.

Recall

Let $\gamma(t) = (\gamma_1(t), \dots, \gamma_n(t))$ for all $t \in (a, b)$ be a parameterized curve of class $\mathcal{C}^2$. We have $\gamma'(t_0) = \lim_{h \rightarrow 0} \frac{\gamma(t_0+h) - \gamma(t_0)}{h}$. When $\gamma'(t_0)$ exists and is nonzero, then we can think of this as the tangent vector of the curve γ at the point $\gamma(t_0)$. Similarly, we can think of $\gamma''(t_0)$ to be the acceleration of γ at $\gamma(t_0)$.

Proposition

Let $u : (a, b) \rightarrow \mathbb{R}^n$ be a $\mathcal{C}^1$ function such that $\|u(t)\| = 1$ for all $t \in (a, b)$. Then $u(t) \cdot u'(t) = 0$.

Proof:

We have $1 = \|u(t)\|^2 = u(t) \cdot u(t) = \sum_{i=1}^n u_i(t)^2$, implying by the product rule

$$2u(t) \cdot u'(t) = 2 \sum_{i=1}^n u_i(t)u'_i(t) = \frac{d}{dt} \sum_{i=1}^n u_i(t)^2 = \frac{d}{dt} 1 = 0 \quad \blacksquare$$

Why do we care? If γ is unit-speed, then the proposition tells us that $\gamma'(t) \cdot \gamma''(t) = 0$, implying $\gamma''(t) = 0$ or $\gamma'(t)$ is perpendicular to $\gamma''(t)$.

Let's now try to measure how much a curve "curves".

Definition

Let $\gamma : (a, b) \rightarrow \mathbb{R}^n$ be a unit-speed curve of class $\mathcal{C}^2$. We define $T(s) := \gamma'(s)$, and is called the unit tangent vector of γ at $\gamma(s)$. We define the curvature of γ at $\gamma(s)$ to be $\kappa(s) := \|\gamma''(s)\| \geq 0$, and at points when $\kappa(s) > 0$, we define the principal unit normal to γ at $\gamma(s)$ to be $N(s) := \frac{\gamma''(s)}{\kappa(s)}$. Note that $\|\gamma''(s)\| = \kappa(s)$, so the acceleration is 0 if and only if the curvature is 0. Therefore we have $\kappa(s) = 0$ or $T(s)$ and $N(s)$ are perpendicular by the above proposition.

Example

Consider the circle $x^2 + y^2 = a^2$, parameterized by arc-length as $\gamma(s) = (a \cos(s/a), a \sin(s/a))$. Then $T(s) = \gamma'(s) = (-\sin(s/a), \cos(s/a))$, and $\gamma''(s) = (-1/a \cos(s/a), -1/a \sin(s/a)) = -\gamma(s)/a^2$, and thus $\kappa(s) = \|\gamma''(s)\| = \frac{1}{a}$ and $N(s) = \gamma''(s)/\kappa(s) = -\gamma(s)/a$.

Example

Consider the helix $\gamma(s) = (a \cos(cs), a \sin(cs), bcs)$ for $c = 1/\sqrt{a^2 + b^2}$ and $s \in \mathbb{R}$. Then $\gamma(s)$ is a unit speed parameterization of the helix, $\gamma'(s) = (-a \sin(cs), a \cos(cs), bc)$, so $\|\gamma'(s)\| = \sqrt{a^2 c^2 + b^2 c^2} = |c| \sqrt{a^2 + b^2} = 1$ for all $s \in \mathbb{R}$. Therefore, $T(s) = \gamma'(s) = (-a \sin(cs), a \cos(cs), bc)$, and $\gamma''(s) = (-ac^2 \cos(cs), -ac^2 \sin(cs), 0)$, implying $\kappa(s) = \|\gamma''(s)\| = ac^2 = \frac{a}{a^2 + b^2} > 0$, and $N(s) = \frac{\gamma''(s)}{\kappa(s)} = -(\cos(cs), \sin(cs), 0)$.

Remark

We see the helix and circle are both constant curvature, but are very different. We'll need another

quantity (adding a dimension) to distinguish between space curves - torsion.

Example

Let's try our definition of curvature on lines, if the definition is at all reasonable, we should get 0! We know a line ℓ in $\mathbb{R}^n$ is completely determined by a point x_0 and a direction vector v . Using x_0 and v , we obtain the parameterization $\gamma(t) = x_0 + tv$, $t \in \mathbb{R}$ of ℓ . Is this unit speed? $\gamma'(t) = v$, so γ is unit speed if and only if $\|v\| = 1$. If $\|v\| = 1$, then $\gamma(s) = x_0 + sv$, $s \in \mathbb{R}$, is a unit speed parameterization of ℓ . Then $\gamma'(s) = v = T(s)$, and $\gamma''(s) = 0$, implying $\kappa(s) = \|\gamma''(s)\| = \|0\| = 0$, so we see the line does not curve.

Proposition

A unit speed curve γ in $\mathbb{R}^n$ is the parameterization of a line if and only if $\kappa(s) = 0$.

Proof:

We just proved the forward direction in the above example. In the reverse direction, γ be a unit speed curve with $\kappa(s) = 0$ for all s , so $\gamma''(s) = 0$ for all s . Integrating, this implies $\gamma'(s) = v$ for all s for some $v \in \mathbb{R}^n$ where v is a unit vector. Integrating again, this implies we have $\gamma(s) = sv + x_0$ for some $x_0 \in \mathbb{R}^n$, which is just the equation of a line in $\mathbb{R}^n$ through the point x_0 with direction v . ■

Remark

For a curve in $\mathbb{R}^2$ with $\kappa(s) > 0$, $N(s)$ is unique up to a sign ($T(s)$ is uniquely determined when it is unit length by the orientation of our parameterization, and then the normal is determined from this), but our convention is that $N(s)$ points "inward", which we can make because we are in $\mathbb{R}^2$. Thus in $\mathbb{R}^3$ this definition fails.

In $\mathbb{R}^n$ for $n > 2$, $N(s)$ is a vector in the plane with normal direction $T(s)$ passing through $\gamma(s)$. The plane spanned by $N(s)$ and $T(s)$ is called the osculating plane.

Chapter 1.2: Binormal vectors, Frenet frames, and torsion

In this chapter, we are working in $\mathbb{R}^3$, where $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^3$ is unit speed (we always use the variable s for a curve parameterized by arc-length).

Recall

The cross product is a function $\times : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ (usually written in operator form), given $u \neq 0, v \neq 0 \in \mathbb{R}^3$, for u and v linearly independent, $u \times v \in \mathbb{R}^3$ is the unique vector satisfying the following three properties:

- (1) $u \times v \perp u$ and $u \times v \perp v$
- (2) $\{u, v, u \times v\}$ is a positively oriented basis in $\mathbb{R}^3$ (this means that it satisfies the right-hand rule)
- (3) $\|u \times v\|$ is the area of the parallelogram spanned by u and v .

This gives us an explicit coordinate description of $u \times v$: write $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$, then (this works for any $u, v \in \mathbb{R}^3$, not just linearly independent):

$$u \times v = \det \begin{bmatrix} e_1 & e_2 & e_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{bmatrix} = (u_2 v_3 - u_3 v_2) e_1 - (u_1 v_3 - v_3 u_1) e_2 + (u_1 v_2 - u_2 v_1) e_3 = (u_2 v_3 - u_3 v_2, v_3 u_1 - u_3 v_1, u_1 v_2 - u_2 v_1)$$

By properties of the determinant (or looking at the explicit coordinates), this is alternating ($u \times v = -v \times u$), as well as bilinear.

Remark

- (1) By looking at the determinant expression above, we have that $u, v \in \mathbb{R}^3$ are linearly dependent if and only if $u \times v = 0$.
- (2) By doing some triangle geometry, one can show $\|u \times v\| = \|u\| \|v\| \sin \theta$, where θ is the angle between u and v . This implies that given orthogonal unit vectors (so $\theta = \pi/2$), $u \times v$ is also a unit vector. In this

case then, this tells us $\{u, v, u \times v\}$ is a positive oriented orthonormal basis.

Definition

Given a unit speed curve $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^3$, we define the binormal vector $B(s) := T(s) \times N(s)$ for all $s \in (\alpha, \beta)$ such that $\kappa(s) > 0$ (and thus $N(s) \neq 0$). By properties of the cross product, this tells us that:

- (1) $\|B(s)\| = 1$ for all s where it is defined, because $\|T(s)\| = \|N(s)\| = 1$
- (2) $\{T(s), N(s), B(s)\}$ is a positively oriented orthonormal basis of $\mathbb{R}^3$ of $\gamma(s)$.

Definition

The coordinate frame $\{T(s), N(s), B(s)\}$ is called the Frenet frame at $\gamma(s)$ (a frame is a function that assigns a basis of $\mathbb{R}^3$ to each point (of the curve)).

Example

Let's look at the helix $\gamma(s) = (a \cos(cs), a \sin(cs), bcs)$ for $s \in \mathbb{R}$, with $a, b > 0$, $c = \frac{1}{\sqrt{a^2 + b^2}}$. Then $T(s) = c(-a \sin(cs), a \cos(cs), b)$, $N(s) = -(\cos(cs), \sin(cs), 0)$, and hence $B(s) = T(s) \times N(s) = c(b \sin(cs), -b \cos(cs), a)$ for all s .

Recall

Let $u, v : (\alpha, \beta) \rightarrow \mathbb{R}^3$ of class $\mathcal{C}^1$. Then

- (1) $\frac{d}{dt}(cu(t)) = cu'(t)$ for all $c \in \mathbb{R}$
- (2) $\frac{d}{dt}(u(t) + v(t)) = u'(t) + v'(t)$
- (3) $\frac{d}{dt}(u(t) \cdot v(t)) = u'(t) \cdot v(t) + u(t) \cdot v'(t)$, and $\frac{d}{dt}(u(t) \times v(t)) = u'(t) \times v(t) + u(t) \times v'(t)$

Recall

A plane in $\mathbb{R}^3$ is completely determined by either:

- (1) A point p_0 and 2 direction vectors v_1, v_2 , in which case we can parameterize the plane as $(x, y, z) = p_0 + sv_1 + tv_2$ for $s, t \in \mathbb{R}$
- (2) A point p_0 and a normal vector n_0 of the plane by $(p - p_0) \cdot n_0 = 0$.

Definition

A parameterized curve $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^3$ is called a planar curve if it is contained in a plane Π in $\mathbb{R}^3$. If Π is parameterized by $(p - p_0) \cdot n_0 = 0$, then for any $s_0 \in (\alpha, \beta)$, $\gamma(s_0)$ has to be on Π . Can choose $p_0 = \gamma(s_0)$ and then for all s , we have $(\gamma(s) - \gamma(s_0)) \cdot n_0 = 0$.

Proposition

A smooth unit speed curve $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^3$ (with $\kappa(s) > 0$ for all s) is a plane curve if and only if there exists a unit vector $n_0 \in \mathbb{R}^3$ such that $B(s) = n_0$ for all $s \in (\alpha, \beta)$ if and only if $\frac{dB}{ds} = 0$ everywhere.

Proof:

In the forward direction, suppose that γ is planar. Therefore, it is contained in a plane Π with unit normal vector n_0 . Then, there exists $s_0 \in (\alpha, \beta)$ such that $(\gamma(s) - \gamma(s_0)) \cdot n_0 = 0$, implying by the product rule that $\gamma'(s) \cdot n_0 = 0$. Differentiating again, we further get $\gamma''(s) \cdot n_0 = 0$. Therefore, $\gamma'(s) = T(s)$ is perpendicular to n_0 , and $\gamma''(s) = N(s)\kappa(s)$ is perpendicular to n_0 (and thus $N(s)$ is perpendicular to n_0). Therefore, since n_0 is a unit vector perpendicular to $N(s)$ and $T(s)$, we have $n_0 = \pm T(s) \times N(s) = B(s)$, so $B(s) = \pm n_0$. But $B(s)$ is a smooth function (since γ is), so $B(s) = n_0$ or $B(s) = -n_0$ for all s , so B is a constant vector. Switching signs if necessary of n_0 , we have the result in this direction.

In the reverse direction, suppose $B(s) = n_0$ for all s , for a fixed unit vector $n_0 \in \mathbb{R}^3$. Because $T(s)$ is perpendicular to $B(s)$, we have $T(s) \cdot B(s) = 0$, which by our above differentiation, shows $\frac{d}{ds}((\gamma(s) - \gamma(s_0)) \cdot n_0) = 0$. Thus $(\gamma(s) - \gamma(s_0)) \cdot n_0$ is constant, and because we may vary s to be equal to s_0 , this shows that the constant is zero. Therefore, by our above definition, we see that γ is planar. ■

Proposition

Let $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^3$ be a $\mathcal{C}^3$ curve with $\kappa(s) > 0$ for all $s \in (\alpha, \beta)$. Then for any $s \in (\alpha, \beta)$, either $\frac{dB}{ds} = 0$

or $\frac{dB}{ds}$ and N are parallel.

Proof:

Since B is a unit vector-valued function, either $\frac{dB}{ds} = 0$ or $\frac{dB}{ds}$ is perpendicular to B . Moreover, since $B = T \times N$, by the product rule, we have

$$\frac{dB}{ds} = \frac{d}{ds}(T \times N) = \frac{dT}{ds} \times N + T \times \frac{dN}{ds} = \kappa(s)N(s) \times N(s) + T \times \frac{dN}{ds} = T \times \frac{dN}{ds}$$

Thus either $\frac{dB}{ds} = 0$, or it is perpendicular to both B and T (the latter by the above cross product), where in the latter case, it is parallel with N (since N is also perpendicular to B and T). ■

Definition

In light of the above proposition, let $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^3$ be a $\mathcal{C}^3$ curve with $\kappa(s) > 0$ for all $s \in (\alpha, \beta)$. Then for any $s \in (\alpha, \beta)$, we have $\frac{dB}{ds}(s) = -\tau(s)N(s)$ for some function $\tau : (\alpha, \beta) \rightarrow \mathbb{R}$, called the torsion of γ .

Remark

The minus sign of τ is so that it follows the right-hand rule, that is, τ is positive when our curve is spiraling “upwards”.

Remark

τ , computed as the (continuous) derivative of the (smooth) cross product of $\mathcal{C}^1$ functions N and T , is continuous, and smooth when γ is smooth.

Remark

τ is identically 0 if and only if we have a plane curve by the second-to-last proposition.

Example

Recall the Helix given by $\gamma(t) = (a \cos(ct), a \sin(ct), bct)$ for $a, b, c > 0$. We have $N(s) = -(\cos(cs), \sin(cs), 0)$, and $B(s) = (bc \sin(cs), -bc \cos(cs), ac)$. Taking the derivative of B , we get that $\frac{dB}{ds} = bc^2(\cos(cs), \sin(cs), 0)$. Therefore, $\tau(s) = bc^2 > 0$.

Remark

So far, we have $\frac{dT}{ds} = \kappa N$, $\frac{dB}{ds} = -\tau N$, what about $\frac{dN}{ds}$? We have $\frac{dN}{ds} = \frac{d}{ds}(B \times T) = \frac{dB}{ds} \times T + B \times \frac{dT}{ds} = -\tau(N \times T) + \kappa(B \times N) = \tau B - \kappa T$. Altogether, these three equations give the linear system of ODEs

$$\begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} T \\ N \\ B \end{bmatrix} = \begin{bmatrix} \frac{dT}{ds} \\ \frac{dN}{ds} \\ \frac{dB}{ds} \end{bmatrix}$$

which is called the **Frenet-Serret** system.

Definition

A rigid motion in $\mathbb{R}^3$ is a rotation followed by a translation, that is, a map $M : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ of the form $M(v) = Av + b$ where $A \in \text{SO}(3)$ and $b \in \mathbb{R}^3$ for all $v \in \mathbb{R}^3$.

Theorem (Fundamental Theorem of Spaces Curves)

Let $\gamma_1, \gamma_2 : (\alpha, \beta) \rightarrow \mathbb{R}^3$ be two $\mathcal{C}^3$ regular unit-speed curves with the same curvature $\kappa(s)$ and torsion $\tau(s)$. Then, there is a rigid motion M such that $\gamma_2(s) = M(\gamma_1(s))$ for all $s \in (\alpha, \beta)$. Moreover, given $\mathcal{C}^3$ functions $\kappa : (\alpha, \beta) \rightarrow \mathbb{R}$ and $\tau : (\alpha, \beta) \rightarrow \mathbb{R}$ with $\kappa(s) > 0$, then there exists a unit-speed curve $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^3$ with curvature $\kappa(s)$ and torsion $\tau(s)$.

Idea of Proof:

The main idea is that the Frenet-Serret system is a system of ODEs, which always has a unique solution

by results from real analysis.

Remark

By the above theorem, we see the following are the only possible cases if we have constant curvature and torsion:

If we have $\kappa = 0$, then we have a line (as seen previously).

If we have $\kappa > 0, \tau = 0$, then we have a circle (we know the circle satisfies this, and the uniqueness of the above theorem gives this).

If we have $\kappa > 0, \tau \neq 0$, then we have a Helix (either spiraling down if negative torsion, and spiraling up if positive torsion).

Definition

Let $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^n$ be a $\mathcal{C}^r$ curve. For all $\alpha_0, \beta_0 \in \mathbb{R}$ satisfying $\alpha < \alpha_0 \leq \beta_0 < \beta$, the arclength of γ from α_0 to β_0 is $\int_{\alpha_0}^{\beta_0} \|\gamma'(t)\| dt$.

Example

Consider the circle $\gamma(t) = (a \sin t, a \cos t)$ for $t \in \mathbb{R}, a > 0$. Then the arclength of the circle from 0 to 2π is

$$\int_0^{2\pi} \|\gamma'(t)\| dt = \int_0^{2\pi} a dt = 2\pi a$$

which agrees with what we know about the circumference of the circle.

Example

Consider the helix $\gamma(t) = (a \cos t, a \sin t, bt)$ for $t \in \mathbb{R}, a, b > 0$. We have $\gamma'(t) = (-a \sin t, a \cos t, b)$. Using this to compute its arclength L from $\gamma(0)$ to $\gamma(2\pi)$, we get that

$$L = \int_0^{2\pi} \|\gamma'(t)\| dt = \int_0^{2\pi} \sqrt{a^2 + b^2} dt = 2\pi \sqrt{a^2 + b^2}$$

Definition

Let $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^n$ be a $\mathcal{C}^r$ curve. The arclength of γ from $t_0 \in (\alpha, \beta)$ is $s(t) = \int_{t_0}^t \|\gamma'(\tau)\| d\tau$.

Example

If γ is unit speed, then $s(t) = \int_{t_0}^t 1 d\tau = t - t_0$, or $t = s + t_0$ (“if you go 1 meter per second, then every second you go 1 meter”).

Parameterizing by arc length

Let $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^n$ be a regular $\mathcal{C}^r$ curve. Then by the Fundamental Theorem of Calculus, $\frac{ds}{dt} = \|\gamma'(t)\| > 0$, so by the 1-dimensional inverse function theorem, there exists a $\mathcal{C}^r$ function $t(s)$ that is an inverse for s , so we can take $\gamma(t) = \gamma(t(s)) = \tilde{\gamma}(s)$, and we have $\|\tilde{\gamma}'(s)\| = 1$, so $\tilde{\gamma}$ is a unit-speed parameterization of γ .

Proposition (Frenet-Serret formulas)

Let $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^3$ be a regular $\mathcal{C}^3$ curve. Then $T = \frac{\gamma'(t)}{\|\gamma'(t)\|}$, $B = \frac{\gamma' \times \gamma''}{\|\gamma' \times \gamma''\|}$, $\kappa = \frac{\|\gamma' \times \gamma''\|}{\|\gamma'\|^3}$, $\tau = \frac{(\gamma' \times \gamma'') \cdot \gamma'''}{\|\gamma' \times \gamma''\|^2}$, and $N = B \times T$ (Alex fill this in later).

Proof:

Tedious algebra. ■

Example

Consider the parabola $\gamma : \mathbb{R} \rightarrow \mathbb{R}^3$ given by $\gamma(t, t^2, 0)$. Then $\gamma'(t) = (1, 2t, 0)$, $\gamma''(t) = (0, 2, 0)$, $\gamma' \times \gamma'' = (0, 0, 2)$, and $\|\gamma' \times \gamma''\| = 2$, and $\|\gamma'(t)\| = \sqrt{1 + 4t^2}$, so its curvature by the above proposition is $\frac{2}{\sqrt{1 + 4t^2}^3}$. Graphing this, this matches our expectation that the parabola’s curvature peaks rapidly at 0.

Suppose $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^n$ be a regular $\mathcal{C}^r$ curve for $r \geq n$. Since γ is regular, by the above, we can always reparameterize it to get a unit speed curve $\tilde{\gamma}(s)$ of class $\mathcal{C}^r$ for $s \in (\tilde{\alpha}, \tilde{\beta})$. Let's try to generalize the construction of the Frenet-Serret system. All we needed was that γ' and γ'' were linearly independent because γ is unit-speed:

Definition

A Frenet curve is a regular curve $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^n$ of class $\mathcal{C}^n$, such that $\{\gamma'(t), \dots, \gamma^{(n-1)}(t)\}$ is linearly independent in $\mathbb{R}^n$ for all $t \in (\alpha, \beta)$.

We want to construct an orthonormal frame out of $\gamma'(t), \dots, \gamma^{(n-1)}(t)$. By reparameterizing if necessary (since γ is regular), we may assume γ is unit-speed. We then apply Gram-Schmidt to $\{\gamma'(s), \dots, \gamma^{(n-1)}(s)\}$: set $e_1 = \gamma' = T$, $e_2 = \frac{\gamma''}{\|\gamma''\|} = N$, which is orthogonal since γ is unit-speed, and nonzero since our frame is linearly independent, and generally after that for $j > 2$,

$$e_j = \frac{\gamma^{(j)} - \sum_{i=1}^{j-1} (\gamma^{(j)} \cdot e_i) e_i}{\|\gamma^{(j)} - \sum_{i=1}^{j-1} (\gamma^{(j)} \cdot e_i) e_i\|}$$

giving us an orthonormal set $\{e_1 = T, e_2 = N, e_3, \dots, e_{n-1}\}$ whose span is an $n-1$ dimensional subspace of $\mathbb{R}^n$, let's denote it by W . We can look at the 1-dimensional subspace $W^\perp$, which has two choices of unit-length vectors that span it (which are negatives of each other). Either one we pick for e_n will give us an orthogonal matrix out of the columns, which will have determinant 1 or -1 . We'll pick the one that gives us a determinant 1 (the other one must give us -1 because it is negative of the previous vector, and changing the sign of a column of a determinant flips the sign of the determinant), and let's call this e_n . This gives us an orthonormal set $\{e_1 = T, e_2 = N, \dots, e_n\}$, and we say it is **positively oriented** because its determinant (taking the vectors as the column) is positive. This is called the **Frenet frame** of γ at $\gamma(s)$.

Theorem

Given the above setup,

$$\begin{bmatrix} e'_1(s) \\ e'_2(s) \\ e'_3(s) \\ \vdots \\ e'_{n-1}(s) \\ e'_n(s) \end{bmatrix} = \begin{bmatrix} 0 & \kappa_1 & 0 & 0 & \dots & 0 \\ -\kappa_1 & 0 & \kappa_2 & 0 & \dots & 0 \\ 0 & -\kappa_2 & 0 & \kappa_3 & \dots & 0 \\ \vdots & & & \ddots & \dots & 0 \\ 0 & \dots & 0 & 0 & 0 & \kappa_{n-1} \\ 0 & \dots & 0 & 0 & -\kappa_{n-1} & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ \vdots \\ e_{n-1} \\ e_n \end{bmatrix}$$

where κ_i is the i th Frenet curvature, defined by $\kappa_i = e'_i \cdot e_{i+1}$, where when $n = 3$, we have $\kappa_1 = \kappa, \kappa_2 = \tau$.

Chapter 1.3: Submanifolds of $\mathbb{R}^2$

Before giving the definition, let us find a sufficient condition for a level curve $C = \{(x, y) \in \mathbb{R}^2 : f(x, y) = k, f : A \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}\} \subseteq \mathbb{R}^2$ to be parameterizable by one variable $t \in (\alpha, \beta)$. We need the implicit function theorem:

Theorem (Implicit Function Theorem in $\mathbb{R}^2$)

Let $f : A \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ be a $\mathcal{C}^k$ function in a neighbourhood of some $(a, b) \in A$ and suppose that $f(a, b) = 0$. Then:

(1) If $\frac{\partial}{\partial y} f(a, b) \neq 0$ then there exists a unique $\mathcal{C}^k$ function $y = g(x)$ on C (the level set above) in a neighbourhood of (a, b) .

(2) If $\frac{\partial}{\partial x} f(a, b) \neq 0$, then there exists a unique $\mathcal{C}^k$ function $x = h(y)$ on C in a neighbourhood of (a, b) .

Remark

This tells us that if $\frac{\partial}{\partial y} f(a, b) \neq 0$, then C looks like $\{(x, f(x))\}$ around (a, b) , so we can parameterize it by $\gamma(t) = (t, g(t))$, and symmetrically for y .

Example

Take $f(x, y) = y - x^2$ on $\mathbb{R}^2$, which is smooth. Then its zero level set is $C = \{(x, y) \in \mathbb{R}^2 : y = x^2\}$ is a parabola in $\mathbb{R}^2$. Since $\frac{\partial}{\partial y} f = 1 \neq 0$ for all points on C , y can be expressed as a unique smooth function of x (in this case, $y = x^2$). This gives us the parameterization $\gamma : \mathbb{R} \rightarrow \mathbb{R}^2$ by $\gamma(t) = (t, t^2)$. Trying the other side, we have $\frac{\partial}{\partial x} f = -2x$, which vanishes at the origin. Thus we can express x as a unique smooth function of y in a neighborhood of any point not including 0, and the theorem fails in a neighborhood of 0. This corresponds to the fact that $x = \pm\sqrt{y}$, and we want to pick one square root. We can however parameterize $(a, b) \in C$ such that $(a, b) \neq 0$ in the two cases $a > 0, a < 0$, where $x = \sqrt{y}$ if $a > 0$, corresponding to a parameterization $\gamma(t) = (\sqrt{t}, t)$, $t > 0$ and $x = -\sqrt{y}$ if $a < 0$, corresponding to a parameterization $\gamma(t) = (-\sqrt{t}, t)$, $t < 0$.

What if $\frac{\partial}{\partial x} f(a, b) = \frac{\partial}{\partial y} f(a, b) = 0$? Then we may still be able to express $y = g(x)$ or $x = h(y)$ at points on C , but it may fail to be smooth or we might lose uniqueness.

Example

Define $f(x, y) = x^3 - y^2$, which is smooth on $\mathbb{R}^2$, and consider the zero level set $C = \{(x, y) \in \mathbb{R}^2 : x^3 = y^2\}$. We have $\frac{\partial}{\partial x} f = 3x^2$ which equals 0 if and only if $x = 0$, and $\frac{\partial}{\partial y} f = -2y$ which again is 0 if and only if $y = 0$. On this curve, we have $x = 0$ or $y = 0$ only at $(x, y) = 0$, so we cannot apply the implicit function theorem in a neighborhood of 0. However, here we can write $x = y^{2/3}$ on the entirety of C , giving us a parameterization $\gamma(t) = (t^{2/3}, t)$, $t \in \mathbb{R}$, but γ is not smooth at $t = 0$.

Example

Define $f(x, y) = x^4 - y^2$, which is smooth on $\mathbb{R}^2$, and consider the zero level set $C = \{(x, y) \in \mathbb{R}^2 : x^4 = y^2\}$. Then $x^4 = y^2$ if and only if $y = x^2$ or $y = -x^2$, implying $C = \{(x, y) \in \mathbb{R}^2 : y = x^2\} \cup \{(x, y) \in \mathbb{R}^2 : y = -x^2\}$, where its parameterization near the origin is clearly not unique (draw a graph), which can be seen by the implicit function theorem not applying since $\frac{\partial}{\partial x} f = 4x^3$, $\frac{\partial}{\partial y} f = -2y$, both of which are 0 only at the origin, so we cannot apply the implicit function theorem at the origin. In this case, C has 2 “branches” passing through the origin, namely the upwards parabola $y = x^2$ and the downwards parabola $y = -x^2$, given by C_1 parameterized by $\gamma(t) = (t, t^2)$, $t \in \mathbb{R}$ and C_2 parameterized by $\gamma(t) = (t, -t^2)$, $t \in \mathbb{R}$.

As a consequence of the implicit function theorem, we get:

Proposition

Let f be a smooth function and (a, b) be a point on the zero level set of f (so $f(a, b) = 0$). If $\nabla f(a, b) \neq 0$, then there exists a smooth regular parameterization of the zero level set of f in a neighborhood of (a, b) .

Proof:

If $\nabla f(a, b) \neq 0$, then $\frac{\partial}{\partial x} f \neq 0$ or $\frac{\partial}{\partial y} f \neq 0$. Without loss of generality, suppose $\frac{\partial}{\partial x} f \neq 0$. Then, by the implicit function theorem, there exists a smooth function $h(y)$ such that $x = h(y)$ on a neighborhood of (a, b) . This gives us a smooth parameterization $\gamma(t) = (h(t), t)$ defined on a sufficiently small interval of the zero level set of f near (a, b) . Then, $\gamma'(t) = (h'(t), 1)$, which does not vanish (because of the second parameter), so γ is regular. ■

We see that many smooth curves we can write locally as the zero level sets of some smooth functions, which is what a (1 dimensional) submanifold is.

Definition

A subset $M \subseteq \mathbb{R}^2$ is called a 1-dimensional (smooth) submanifold (abbreviated 1-submanifold) if for all $p \in M$, there exists an open ball B centered at p and a smooth function $f : B \rightarrow \mathbb{R}$ such that $M \cap B = \{(x, y) \in M : f(x, y) = 0\} = f^{-1}(0)$ and $\nabla f(x, y) \neq 0$ for all $(x, y) \in M \cap B$.

Example

Suppose $f(x, y) = y - x^2$ for $(x, y) \in \mathbb{R}^2$. Then its zero level set is the graph $C = \{(x, y) \in \mathbb{R}^2 : f(x, y) = 0\} = \{(x, y) \in \mathbb{R}^2 : y = x^2\}$. We have $\frac{\partial}{\partial y}f = 1 \neq 0$ globally, and so $\nabla f(x, y) \neq 0$ globally, so C is a 1-submanifold (with function f suitably restricted for any point (a, b) and any neighborhood of it). This corresponds (by the implicit function theorem) to a global regular parameterization $\gamma(t) = (t, t^2)$ of C .

Example

Suppose $f(x, y) = x^2 + y^2 - 1$, $(x, y) \in \mathbb{R}^2$. Then take $C = \{(x, y) \in \mathbb{R}^2 : f(x, y) = 0\} = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$. Then $\frac{\partial}{\partial y}f = 2y = 0$ if and only if $y = 0$ which occurs only at $(\pm 1, 0)$ within C , and $\frac{\partial}{\partial x}f = -2x = 0$ if and only if $x = 0$ which occurs only at $(0, \pm 1)$ within C , so $\nabla f(x, y) \neq 0$ globally, so C is a 1-submanifold with function f (suitably restricted) for any point (a, b) and any neighborhood. However, because the x and y coordinates do vanish at times, we cannot use the implicit function theorem to get one global parameterization, we need to split up our analysis into pieces. Specifically, for $(a, b) \neq (\pm 1, 0)$, by the implicit function theorem we can write $y = g(x)$ (specifically $y = \pm\sqrt{1 - x^2}$). Thus, for $a \neq \pm 1, b > 0$, we can parameterize this piece of the circle by $y = \sqrt{1 - x^2}$, $-1 < x < 1$, corresponding to parameterization $\gamma_1(t) = (t, \sqrt{1 - t^2})$, $t \in (-1, 1)$, and for $a \neq \pm 1, b < 0$, we can parameterize this piece by $y = -\sqrt{1 - x^2}$, corresponding to parameterization $\gamma_2(t) = (t, -\sqrt{1 - t^2})$, $t \in (-1, 1)$, and for $(a, b) \neq (0, \pm 1)$, we can similarly parameterize it by functions $x = \pm\sqrt{1 - y^2}$, $-1 < y < 1$, corresponding to parameterizations $\gamma_3(t) = (\sqrt{1 - s^2}, s)$, $s \in (-1, 1)$ and $\gamma_4(t) = (-\sqrt{1 - s^2}, s)$, $s \in (-1, 1)$.

Remarks

Consider the above example.

- (1) These four parameterization pieces collectively cover the whole circle (each gives an open semicircle as its image), so every point is described by at least one of the regular parameterizations $\gamma_1, \gamma_2, \gamma_3, \gamma_4$.
- (2) Each parameterization is a homeomorphism (continuous bijection with continuous inverse) from $(-1, 1)$ onto an open semicircle. Such an inverse is of the form $\gamma_1^{-1}(x, y) = x$, which is clearly continuous.
- (3) Consider the intersection of two parameterization's images, say $\text{im } \gamma_1 \cap \text{im } \gamma_3$, which is the quadrant 1 open quarter-circle. Then for some (x, y) in this intersection, we have $(t, \sqrt{1 - t^2}) = (x, y) = (\sqrt{1 - s^2}, s)$, and so $t = \sqrt{1 - s^2}$ for $s \in (0, 1)$. Write $t = \phi(s) = \sqrt{1 - s^2}$. Then we have $\gamma_3(s) = (\gamma_1 \circ \phi)(s)$ for $s \in (0, 1)$, so we see γ_1 and γ_3 are related by the map $\phi : (0, 1) \rightarrow (0, 1)$, defined by $\phi(s) = \sqrt{1 - s^2}$, which is a diffeomorphism (a smooth bijection with smooth inverse, where its inverse is $t \mapsto \sqrt{1 - t^2}$). We can equivalently write $\phi = \gamma_1^{-1} \circ \gamma_3 : (0, 1) \rightarrow (0, 1)$. Such a function is called a **transition function**, which corresponds to the “change of variables” between the two parameterizations which parameterize our curve.

Putting these three points together, we get an equivalent way to think about 1-submanifolds:

Given a 1-submanifold M , by the implicit function theorem (see the most recent proposition) and the above work, every point $p \in M$ is in the image of a smooth regular parameterization $\gamma : (\alpha, \beta) \rightarrow M$ such that γ is a smooth bijection onto its image $\text{im } \gamma = \gamma((\alpha, \beta))$ with continuous inverse. It is a bijection onto its image by the implicit function theorem, and has smooth inverse that just inverts the “time” coordinate. We can generalize this:

Definition

Let M be a 1-submanifold. An atlas for M is a collection of smooth regular parameterizations that are homeomorphisms onto their images, such that the union of their images is M , and if $\gamma_s, \gamma_{s'}$ are two of these parameterizations whose images have a nonempty intersection, then the “transition function”

$\phi_{s,s'} = \gamma_s^{-1} \circ \gamma_{s'}$ is a diffeomorphism. Each of these parameterizations is called a coordinate chart (or coordinate patch).

Example

Suppose $M = \{(x, y) : y = x^2\} \subseteq \mathbb{R}^2$. Then $\gamma(t) = (t, t^2), t \in \mathbb{R}$ is a chart with image M , so $\{\gamma : \mathbb{R} \rightarrow M\}$ is an atlas of M that has a single chart!

Remark

In general, the graph of a smooth (scalar) function $f : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$, that is, $M = \{(x, y) \in \mathbb{R}^2 : x \in I, y = f(x)\}$, is a 1-submanifold of $\mathbb{R}^2$ that admits the atlas with one chart consisting of $\{\gamma : I \rightarrow M, t \mapsto (t, f(t))\}$, which only has one chart.

Example

Take the circle $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$. The above shows you cannot make an atlas for a circle with 1 chart, since the charts are homeomorphisms and so must take their non-compact (open) domains to non-compact images, and the circle is compact. Our previous examples showed that M is a 1-submanifold for $\mathbb{R}^2$, the example we constructed previously had 4 charts. On the assignments, using stereographic projection, we will construct one with only two charts. This is done by taking any point $(x, y) \neq (0, 1)$, and drawing the line segment from (x, y) to $(0, 1)$ then extending it to the x -axis. By analytic geometry, this point on the x -axis has x -coordinate $\frac{x}{1-y}$: the inverse map of this will be our chart, which is $t \mapsto \left(\frac{2t}{t^2+1}, \frac{t^2-1}{t^2+1}\right)$. We can do the same thing with the south pole to get the second chart to cover the north pole, where the point on the x -axis has x coordinate $\frac{x}{1+y}$, which gives the chart $t \mapsto \left(\frac{2t}{t^2+1}, -\frac{t^2-1}{t^2+1}\right)$.

Chapter 2: Surfaces

Definition

A subset S of $\mathbb{R}^3$ is a surface if, for every point $p \in S$, there is an open disk $U \subseteq \mathbb{R}^2$ and an open ball W in $\mathbb{R}^3$ containing p such that U is homeomorphic to $S \cap W$. That is, there exists a continuous map with continuous inverse $\sigma : U \subseteq \mathbb{R}^2 \rightarrow S \cap W \subseteq \mathbb{R}^3$ given by $(s, t) \mapsto (\sigma_1(s, t), \sigma_2(s, t), \sigma_3(s, t))$. Each homeomorphism σ is called a surface patch, or a parameterization, or coordinate patch. The collection of all patches is called an atlas. Each point $p \in S$ is contained in at least one patch of the atlas. Finally, if $\sigma : U \rightarrow S \cap W$ and $\tilde{\sigma} : \tilde{U} \rightarrow S \cap \tilde{W}$ are any two patches whose images overlap, then we call $\sigma^{-1} \circ \tilde{\sigma} : \tilde{\sigma}^{-1}(\text{im } \sigma \cap \text{im } \tilde{\sigma}) \subseteq \mathbb{R}^2 \rightarrow \sigma^{-1}(\text{im } \sigma \cap \text{im } \tilde{\sigma}) \subseteq \mathbb{R}^2$ the transition function between σ and $\tilde{\sigma}$, which is a homeomorphism since compositions of homeomorphisms are homeomorphisms.

We can state a more general definition using open sets:

Definition

In $\mathbb{R}^n$, an open set is a subset $U \subseteq \mathbb{R}^n$ that is a union of open balls. In practice, U is a subset of $\mathbb{R}^n$ described by strict inequalities or all of $\mathbb{R}^n$.

Example

- (1) $\mathbb{R}^n = \bigcup_{x \in \mathbb{R}^n} B_r(x)$ for any $r > 0$, so $\mathbb{R}^n$ is open.
- (2) Open balls are open.
- (3) The half plane $U = \{(x, y) \in \mathbb{R}^2 : y > 0\} = \bigcup_{(x, y) \in U} B_{|y|/2}(x, y)$ is open.

Remark

In the above definition of a surface, we can replace the open ball centered at p with an open set containing p .

Example

A plane Π is determined by a point $p_0 \in \mathbb{R}^3$ and two direction vectors $v_1, v_2 \in \mathbb{R}^3$. For every point $p \in \Pi$, we have $p = p_0 + sv_1 + tv_2$ for some $s, t \in \mathbb{R}$. We obtain the parameterization $\sigma : \mathbb{R}^2 \rightarrow \Pi$ determined by $\sigma(s, t) = p_0 + sv_1 + tv_2$ for all $(s, t) \in \mathbb{R}^2$. This gives us concrete coordinates if we write $p_0 = (x_0, y_0, z_0)$, $v_1 = (a_1, b_1, c_1)$, $v_2 = (a_2, b_2, c_2)$, giving $\sigma(s, t) = (x_0 + sa_1 + ta_2, y_0 + sb_1 + tb_2, z_0 + sc_1 + tc_2)$. Such a σ is a homeomorphism (the inverse just gives back s, t from a point $p_0 + sv_1 + tv_2$), though continuity of the inverse is a little annoying to check. It is easier to describe Π as an equation of the form $ax + by + cz = d$ where $n := (a, b, c) \neq 0$ is normal to Π . Since $n \neq 0$, its coordinates are not all zero, so say it is a . Then we can write $x = \frac{d-by-cz}{a}$, so define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $(y, z) \mapsto (d - by - cz)/a$, and we see Π is the graph of $f : \Pi = \{(x, y, z) \in \mathbb{R}^3 : x = f(y, z)\}$, giving a parameterization $\sigma : \mathbb{R}^2 \rightarrow \Pi$ by $(u, v) \mapsto (f(u, v), u, v)$, where this being a homeomorphism is a lot easier to verify since its inverse is $(x, y, z) \mapsto (y, z)$.

Example

Take a continuous function $f : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$, U open. Then the graph of f , $S = \{(x, y, z) \in \mathbb{R}^3 : (x, y) \in U, z = f(x, y)\}$ is a surface with the natural parameterization $\sigma : U \rightarrow S$, $\sigma(x, y) = (x, y, f(x, y))$ for $(x, y) \in U$. We take in the above definition $W = \mathbb{R}^3$, where σ is continuous because f is, and this is by construction bijective, with inverse $(x, y, z) \mapsto (x, y)$, which is continuous as a coordinate projection. Thus σ is a homeomorphism. This gives an atlas of S with a single coordinate patch, and the same construction works with 2 variable functions of xz and of yz .

Example

Consider a sphere S^2 of radius a centered at the origin, $x^2 + y^2 + z^2 = a^2$. We can solve $z = \pm\sqrt{a^2 - x^2 - y^2}$, $y = \pm\sqrt{a^2 - x^2 - z^2}$, $x = \pm\sqrt{a^2 - y^2 - z^2}$ with appropriate restrictions of x, y, z . Consider $z = \pm\sqrt{a^2 - x^2 - y^2}$, which is only valid for (x, y) in the (interior of the) projection of S^2 onto the xy -plane, which is the disk $x^2 + y^2 < a^2$, call it U , where we take the interior to ensure we get (x, y) in an open set in $\mathbb{R}^2$. This gives us a parameterization $\sigma_1 : U \rightarrow S_1 = \{(x, y, z) \in \mathbb{R}^3 : z = \sqrt{a^2 - x^2 - y^2}, (x, y) \in U\}$ by $\sigma_1(u, v) = (u, v, \sqrt{a^2 - u^2 - v^2})$ on the upper-half sphere, and another parameterization $\sigma_2 : U \rightarrow S_2 = \{(x, y, z) \in \mathbb{R}^3 : z = -\sqrt{a^2 - x^2 - y^2}, (x, y) \in U\}$ by $\sigma_2(u, v) = (u, v, -\sqrt{a^2 - u^2 - v^2})$ on the lower half sphere. Since all of the S_i we get from these parameterizations are graphs of functions, where the parameterizations are the parameterizations of graphs we constructed above, we are guaranteed they are homeomorphisms. We get the other parameterizations similarly, $\sigma_3 : U \rightarrow S_3 = \{(x, y, z) \in \mathbb{R}^3 : y = \sqrt{a^2 - x^2 - z^2}, (x, z) \in U\}$, given by $\sigma_3(u, v) = (u, \sqrt{a^2 - u^2 - v^2}, v)$, and define $\sigma_4, \sigma_5, \sigma_6$ similarly, to get an atlas $\{\sigma_1, \dots, \sigma_6\}$ for S^2 .

Example

We'll now see an example of something that is not a surface. Consider the double-cone C given by $x^2 + y^2 = z^2$, which as the name suggests, looks like an upside-down cone on top of another cone, with the vertices connected. The problem is at the vertex at the origin. For a contradiction, suppose there exists a homeomorphism $\sigma : U \rightarrow C \cap W$, where U is open in $\mathbb{R}^2$ and contains the preimage of the origin, call it p , and W is open in $\mathbb{R}^3$ and contains the origin. By shrinking U if necessary, we may assume U is an open disk centered at p . Now consider deleting p and the origin from U and W respectively. $U \setminus \{p\}$ is a punctured disk, which is connected, and thus $\sigma(U \setminus \{p\})$ is connected since σ is a homeomorphism, but $\sigma(U \setminus \{p\}) = W \setminus \{0\}$ which is a piece of the cone on top and a piece of the cone on the bottom, which is disconnected since we removed the vertex where they touch (the origin), a contradiction.

Definition

Recall that given a regular parameterized curve $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^n$, for all $t_0 \in (\alpha, \beta)$, $\gamma'(t_0)$ is tangent to γ at $\gamma(t_0)$, so that we have a well defined tangent line at $\gamma(t_0)$: $L = \{\gamma(t_0) + s\gamma'(t_0), s \in \mathbb{R}\}$. Moreover, if $M \subseteq \mathbb{R}^2$ is a smooth 1 dimensional submanifold of $\mathbb{R}^2$, for all $p \in M$, there exists an open disc D centered at p and a smooth regular parameterization $\gamma : (\alpha, \beta) \rightarrow M \cap D$ that is also a homeomorphism, and we can use the tangent line above to define the tangent line $T_p M$ to M at $p = \gamma(t_0)$:

$$T_p M := \{(\gamma(t_0) + s\gamma'(t_0)) : s \in \mathbb{R}\}$$

We can move towards doing the same for surfaces, but we need to expand our definition to include smoothness:

Definition

A coordinate chart $\sigma : U \rightarrow S \cap W$ of a surface $S \subseteq \mathbb{R}^3$ (U open in $\mathbb{R}^2$, W open in $\mathbb{R}^3$) is regular if it is smooth C^∞ and the partial derivatives $\sigma_u(u, v)$ and $\sigma_v(u, v)$ are linearly independent for all $(u, v) \in U$.

Remark

In practice, we check the second condition by verifying $\sigma_u(u, v) \times \sigma_v(u, v) \neq 0$ for all $(u, v) \in U$.

Definition

A smooth (or regular) surface in $\mathbb{R}^3$ is a surface that admits an atlas whose coordinate charts are all regular.

Example

The graph of a smooth function $f : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ (U open) is a smooth surface, $G = \{(x, y, z) \in U \times \mathbb{R} : z = f(x, y)\}$ with the chart we've seen previously: $\sigma : U \rightarrow G$ by $\sigma(u, v) = (u, v, f(u, v))$ for $(u, v) \in U$. This is smooth since f is smooth. We have $\sigma_u = (1, 0, f_u)$, $\sigma_v = (0, 1, f_v)$, which looking at the first two coordinates is clearly linearly independent, so G is a smooth surface by definition.

Example

Consider the sphere S^2 defined by $x^2 + y^2 + z^2 = 1$, with spherical coordinates $\sigma(\theta, \phi) = (\cos \theta \sin \phi, \sin \theta \sin \phi, \cos \phi)$ (θ is angle from x axis in xy -plane, ϕ is angle from z axis) defined on $(\theta, \phi) \in (0, 2\pi) \times (0, \pi)$, which is a chart for the sphere minus the arc $\theta = 0$. σ is a homeomorphism (it is clearly smooth and bijective, essentially by definition of spherical coordinates), though we'll skip proving the inverse is continuous because machinery we'll prove in the future will do that for us. Its partial derivatives are $\sigma_\theta = (-\sin \theta \sin \phi, \cos \theta \sin \phi, 0)$ and $\sigma_\phi = (\cos \theta \cos \phi, \sin \theta \cos \phi, -\sin \phi)$, and the cross product is

$$\sigma_u \times \sigma_v = (-\cos \theta \sin^2 \phi, -\sin \theta \sin^2 \phi, -\cos \phi \sin \phi)$$

Example

Consider a smooth function $f : U \rightarrow \mathbb{R}$, $U \subseteq \mathbb{R}^3$ an open subset, such that $\nabla f(a, b, c) \neq 0$ for all $(a, b, c) \in S$ where $S = \{(x, y, z) \in \mathbb{R}^3 : f(x, y, z) = 0\}$. Then S is a smooth surface. This follows by the implicit function theorem, since $\nabla f(a, b, c) \neq 0$ so one of $\frac{\partial}{\partial x} f, \frac{\partial}{\partial y} f, \frac{\partial}{\partial z} f \neq 0$, say with respect to x . Then the implicit function theorem tells us in a neighborhood of (a, b, c) , there is a unique smooth scalar function $g : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $x = g(y, z)$ in a neighbourhood of (a, b, c) , and so we can get a regular smooth parameterization $(u, v) \mapsto (g(u, v), u, v)$ in this neighbourhood. We can get a regular chart around each point in this way, giving that S is a smooth surface.

Example

The above example is a useful result! It allows to check many things are smooth surfaces without writing down charts. For example, the sphere of radius a , $S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = a^2\}$ works with the above example with $f(x, y, z) = x^2 + y^2 + z^2 - a^2$. Then $\nabla f(x, y, z) = (2x, 2y, 2z)$ which equals 0 if and only if $(x, y, z) = 0$ which is not a point on S^2 .

Example

If $\nabla f(a, b, c) = 0$ in our above example, we need to look further: Consider the double cone $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2\}$ which is not a surface. Writing $f(x, y, z) = x^2 + y^2 - z^2$, we see f is smooth but $\nabla f(x, y, z) = (2x, 2y, -2z)$ which is 0 if and only if $(x, y, z) = 0$, but this is a point on S ! Otherwise, if we delete the origin from S , this example works to show the union of the vertexless cones $S_+ \cup S_- = \{z = \sqrt{x^2 + y^2} : (x, y) \neq 0\} \cup \{z = -\sqrt{x^2 + y^2} : (x, y) \neq 0\}$ is a smooth surface.

Definition

A 2-dimensional smooth submanifold of $\mathbb{R}^3$ (abbreviated 2-submanifold) is a subset $M \subseteq \mathbb{R}^3$ such that

for all $p \in M$, there exists an open subset $W \subseteq \mathbb{R}^3$ containing p such that $M \cap W = \{(x, y, z) \in W : f(x, y, z) = 0\}$ for some smooth function $f : W \rightarrow \mathbb{R}$ such that $\nabla f(p) \neq 0$.

Remark

A 2-dimensional smooth submanifold of $\mathbb{R}^3$ is a smooth surface because it is the union of graphs of smooth functions by the implicit function theorem, which are parameterizable by regular parameterizations, which gives us our coordinate charts.

The converse of the above remark is also true. We can right away see that the graph of a smooth 2-variable scalar function (which is a smooth surface as seen previously) is a 2-dimensional smooth submanifold. Indeed, suppose we have a smooth function $g : U \rightarrow \mathbb{R}$ with $U \subseteq \mathbb{R}^2$ open. Then its graph $\{(x, y, z) \in \mathbb{R}^3 : z = g(x, y), (x, y) \in U\}$ can be written as $\{(x, y, z) \in U \times g(U) : f(x, y, z) = 0\}$ for the smooth scalar function $f(x, y, z) = z - g(x, y)$ for all $(x, y, z) \in U \times g(U)$, where we see $\nabla f(x, y, z) = (-\frac{\partial}{\partial x}g(x, y), -\frac{\partial}{\partial y}g(x, y), 1) \neq 0$ globally. And thus we see the graph of g is a 2-submanifold. To prove the full converse (note for anyone reading my notes, I put this as a theorem to see it later, Ruxandra did not):

Theorem

Let $S \subseteq \mathbb{R}^3$ be a smooth surface. Then S is locally the graph of a smooth 2-variable scalar function.

Proof:

Let $p \in S$. Then there exists an open set $W \subseteq \mathbb{R}^3$ and an open set $U \subseteq \mathbb{R}^2$ and a smooth homeomorphism $\sigma : U \rightarrow S \cap W$, with $\sigma(u, v) = (\sigma_1(u, v), \sigma_2(u, v), \sigma_3(u, v))$ such that σ_u and σ_v are linearly independent on U . Therefore, the Jacobian $D\sigma = [\sigma_u \mid \sigma_v]$ has rank 2 everywhere on U . Let $(u_o, v_o) \in U$. After possibly permuting the coordinates in $\mathbb{R}^3$, we can assume that the first two rows of $D\sigma(u_o, v_o)$ are linearly independent. Thus $\begin{bmatrix} \sigma_{1,u}(u_o, v_o) & \sigma_{1,v}(u_o, v_o) \\ \sigma_{2,u}(u_o, v_o) & \sigma_{2,v}(u_o, v_o) \end{bmatrix}$ has rank 2. Let $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the projection onto the first two coordinates, $(x, y, z) \mapsto (x, y)$. Then taking the derivative of $\pi \circ \sigma : U \rightarrow \mathbb{R}^2$ $(u, v) \mapsto (\sigma_1(u, v), \sigma_2(u, v))$, we have $D(\pi \circ \sigma)(u_o, v_o) = \begin{bmatrix} \sigma_{1,u}(u_o, v_o) & \sigma_{1,v}(u_o, v_o) \\ \sigma_{2,u}(u_o, v_o) & \sigma_{2,v}(u_o, v_o) \end{bmatrix}$, which we see is our matrix above, so it has rank 2, and thus is invertible. So by the Inverse Function Theorem, the map $\pi \circ \sigma$ is locally invertible at (u_o, v_o) : There exists an open set $\tilde{U} \subseteq \mathbb{R}^2$ containing (u_o, v_o) and an open set $\tilde{V} \subseteq \mathbb{R}^2$ containing $(\pi \circ \sigma)(u_o, v_o)$ and a smooth map $\Phi : \tilde{V} \rightarrow \tilde{U}$ such that Φ is the inverse of $\pi \circ \sigma$. Now, define $\tilde{\sigma} := \sigma \circ \Phi : \tilde{V} \rightarrow S$, which in coordinates is $(\tilde{u}, \tilde{v}) \mapsto (\sigma_1(\Phi(\tilde{u}, \tilde{v})), \sigma_2(\Phi(\tilde{u}, \tilde{v})), \sigma_3(\Phi(\tilde{u}, \tilde{v})))$. But $(\sigma_1(\Phi(\tilde{u}, \tilde{v})), \sigma_2(\Phi(\tilde{u}, \tilde{v}))) = (\pi \circ \sigma)(\Phi(\tilde{u}, \tilde{v})) = (\pi \circ \sigma) \circ \Phi(\tilde{u}, \tilde{v}) = (\tilde{u}, \tilde{v})$ because Φ is the inverse of $\pi \circ \sigma$, and so $\tilde{\sigma}(\tilde{u}, \tilde{v}) = (\tilde{u}, \tilde{v}, (\sigma_3 \circ \Phi)(\tilde{u}, \tilde{v}))$ for all $(\tilde{u}, \tilde{v}) \in \tilde{V}$. Therefore, define $g : \tilde{V} \rightarrow \mathbb{R}$ by $g := \sigma_3 \circ \Phi$, which is smooth by construction of σ_3 and Φ , and we by construction we have $\tilde{\sigma}(\tilde{u}, \tilde{v}) = (\tilde{u}, \tilde{v}, g(\tilde{u}, \tilde{v}))$, and so $\tilde{\sigma}$ is a regular parameterization of the graph of g , which gives the result. ■

Remark

In the proof, we did not use the fact that σ was a homeomorphism, only that it was a smooth bijection. We had $\tilde{\sigma} = \sigma \circ \Phi$ with Φ a diffeomorphism, and $\tilde{\sigma}(\tilde{u}, \tilde{v}) = (\tilde{u}, \tilde{v}, g(\tilde{u}, \tilde{v}))$. By its coordinate representation, we see $\tilde{\sigma}^{-1}(x, y, z) = (x, y)$ is smooth, and thus since $\sigma^{-1} = \Phi \circ \tilde{\sigma}^{-1}$ is a composition of smooth functions, it is smooth for free. Therefore, if we already know that S is a smooth surface and $\sigma : U \rightarrow S \cap W$ is a smooth bijection with $U \subseteq \mathbb{R}^2, W \subseteq \mathbb{R}^3$ open with linearly independent partial derivatives, or equivalently, $D\sigma$ has rank 2, then σ^{-1} is automatically smooth, and in particular continuous, and so for any $p \in S \cap W$, this is a (regular) chart in a neighbourhood of p .

All-in-all, we have:

Theorem

Let $S \subseteq \mathbb{R}^3$. We've seen the following are equivalent:

(1) S is a smooth surface, that is, for all $p \in S$, there exist open sets $U \subseteq \mathbb{R}^2, W \subseteq \mathbb{R}^3$ with $p \in W$ such that there is a smooth homeomorphism $\sigma : U \rightarrow S \cap W$ with coordinates u, v such that σ_u and σ_v are

linearly independent on U (it is regular).

(2) S is a smooth 2-dimensional submanifold of $\mathbb{R}^3$, that is, for all $p \in S$, there exists $W \subseteq \mathbb{R}^3$ open with $p \in W$ and a smooth function $f : W \rightarrow \mathbb{R}$ such that $S \cap W = \{(x, y, z) \in W : f(x, y, z) = 0\}$ and $\nabla f \neq 0$ on $S \cap W$.

(3) For all $p \in S$, there exists an open set $W \subseteq \mathbb{R}^3$ with $p \in W$ and an open set $U \subseteq \mathbb{R}^2$ such that there exists a scalar function $g : U \rightarrow \mathbb{R}$ such that $S \cap W$ is the graph of g . ■

Example

This allows us to easily verify several objects are 2-submanifolds: Suppose Π is a plane described by equation $ax + by + cz = d$ for $(a, b, c) \neq 0 \in \mathbb{R}^3$. Defining the smooth function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ by $f(x, y, z) = ax + by + cz - d$, its zero set is Π and its gradient is $\nabla f(x, y, z) = (a, b, c) \neq 0$ everywhere, therefore Π is a smooth surface. We can parameterize it as well: Take $p_0 \in \Pi$ and direction vectors $w_1, w_2 \in \mathbb{R}^3$, then set $\sigma : \mathbb{R}^2 \rightarrow \Pi \cap \mathbb{R}^3 = \Pi$ by $\sigma(u, v) = p_0 + uw_1 + vw_2$, which is by construction a smooth bijection onto Π , with $D\sigma(u, v) = [w_1 | w_2]$ which is rank 2 since w_1, w_2 are linearly independent as two direction vectors of Π . Therefore, by the remark above, this is a (regular) chart in any neighbourhood of $p \in \Pi$.

Example

The cylinder $S := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$ is the zero set of $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ given by $f(x, y, z) = x^2 + y^2 - 1$, which is smooth and $\nabla f(x, y, z) = (2x, 2y, 0)$ which is zero only at (x, y, z) such that $x = y = 0$, that is, the z -axis, which is not on the cylinder. Therefore $\nabla f(x, y, z) \neq 0$ on S . Let us construct a regular chart for S . Define $\sigma : (0, 2\pi) \times \mathbb{R} \rightarrow S \setminus \{(x, 0, z) \in S : x > 0\}$ by $\sigma(\theta, t) = (\cos \theta, \sin \theta, t)$, which

is by construction a smooth bijection. Then $D\sigma = \begin{bmatrix} \sin \theta & 0 \\ -\cos \theta & 0 \\ 0 & 1 \end{bmatrix}$ which has rank 2 because of the third

row and $\sin \theta, \cos \theta$ never have common zeros. Therefore, σ^{-1} is automatically smooth, so σ is a regular chart. To get an atlas, also use $\tilde{\sigma} : (-\pi, \pi) \times \mathbb{R} \rightarrow S \setminus \{(x, 0, z) \in S : x < 0\}$ by $\tilde{\sigma}(\theta, t) = (\cos \theta, \sin \theta, t)$, and similar ones for the parts of S we excluded.

Definition

Let S be a smooth surface with $p \in S$. Let $\sigma : U \rightarrow S \cap W$ be a regular chart ($U \subseteq \mathbb{R}^2, W \subseteq \mathbb{R}^3$ open) containing p in its image, with coordinates (u, v) . Therefore, there exist $(u_0, v_0) \in U$ such that $p = \sigma(u_0, v_0)$. We define $T_p S$, the tangent plane (space) to S at p , to be the plane in $\mathbb{R}^3$ passing through p with direction vectors $\sigma_u(u_0, v_0)$ and $\sigma_v(u_0, v_0)$. We define the standard unit normal vector of $T_p S$ to be $N_\sigma(u_0, v_0) := \frac{\sigma_u(u_0, v_0) \times \sigma_v(u_0, v_0)}{\|\sigma_u(u_0, v_0) \times \sigma_v(u_0, v_0)\|}$.

Proposition

Let S be a smooth surface, and let $p_0 \in S$.

(1) $T_{p_0} S$ is exactly the set of tangent vectors of all smooth regular curves on S through p_0 at p_0 .

(2) $T_{p_0} S$ is independent of the choice of chart σ .

(3) Up to sign, the N_σ of $T_{p_0} S$ is independent of the choice of chart σ .

Proof:

(1) Let $\gamma : (\alpha, \beta) \rightarrow S$ be a smooth regular curve such that $p_0 = \gamma(t_0)$ for some $t_0 \in (\alpha, \beta)$. Let us check that $\gamma'(t_0) \in \text{Span}\{\sigma_u(u_0, v_0), \sigma_v(u_0, v_0)\}$ which will imply that $\gamma'(t_0) \in T_{p_0} S$. Geometrically, we can use the coordinate chart σ to describe γ in terms of sigma, that is, in a neighbourhood of t_0 , we have $\gamma = \sigma \circ (\sigma^{-1} \circ \gamma)$. Thus, set $\tilde{\gamma} := \sigma^{-1} \circ \gamma : (\tilde{\alpha}, \tilde{\beta}) \rightarrow U$, where $(\tilde{\alpha}, \tilde{\beta})$ is an appropriately small neighborhood of t_0 , and $\tilde{\gamma}$ is smooth because σ is a diffeomorphism and γ is smooth, so writing the coordinate functions $\tilde{\gamma}$ as $\tilde{\gamma}_1, \tilde{\gamma}_2$, we have $\gamma(t) = \sigma(\tilde{\gamma}_1(t), \tilde{\gamma}_2(t))$, and so for t in a neighbourhood of p_0 ,

$$\gamma'(t) = D\sigma(\tilde{\gamma}(t))\tilde{\gamma}'(t) = \sigma_u(\tilde{\gamma}_1(t), \tilde{\gamma}_2(t))\tilde{\gamma}'_1(t) + \sigma_v(\tilde{\gamma}_1(t), \tilde{\gamma}_2(t))\tilde{\gamma}'_2(t)$$

Therefore, since

$$\gamma(t_0) = p_0 = \sigma(u_0, v_0)$$

we have

$$\gamma'(t_0) = \sigma_u(u_0, v_0) \overbrace{\tilde{\gamma}'_1(t_0)}^{\in \mathbb{R}} + \sigma_v(u_0, v_0) \overbrace{\tilde{\gamma}'_2(t_0)}^{\in \mathbb{R}}$$

and so $\gamma'(t_0) \in \text{Span}\{\sigma_u(u_0, v_0), \sigma_v(u_0, v_0)\}$.

Conversely, suppose $v = a\sigma_u(u_0, v_0) + b\sigma_v(u_0, v_0)$ for some $a, b \in \mathbb{R}$. We want a curve $\gamma : (\alpha, \beta) \rightarrow S$ such that $\gamma(t_0) = p_0$ and $\gamma'(t_0) = a\sigma_u(u_0, v_0) + b\sigma_v(u_0, v_0)$ at some $t_0 \in (\alpha, \beta)$. By the above proof, such a curve must be of the form $\gamma(t) = \sigma(\tilde{\gamma}_1(t), \tilde{\gamma}_2(t))$ for some smooth curve $\tilde{\gamma} : (\tilde{\alpha}, \tilde{\beta}) \rightarrow U$ satisfying $\tilde{\gamma}'_1(t_0) = a, \tilde{\gamma}'_2(t_0) = b$. Thus take $\tilde{\gamma}(t) = (u_0, v_0) + (t - t_0)(a, b)$, where this clearly has the desired properties, and take $\gamma = \sigma \circ \tilde{\gamma}$ on a neighbourhood of t_0 .

(2) Let us show that if $\tilde{\sigma} : \tilde{U} \rightarrow S \cap \tilde{W}$ is a chart for p_0 , with coordinates $(\tilde{u}, \tilde{v})$ and $\sigma^{-1}(p_0) = (u_0, v_0) \in U$, $\tilde{\sigma}^{-1}(p_0) = (\tilde{u}_0, \tilde{v}_0) \in \tilde{U}$, then $\text{Span}\{\tilde{\sigma}_u(\tilde{u}_0, \tilde{v}_0), \tilde{\sigma}_v(\tilde{u}_0, \tilde{v}_0)\} = \text{Span}\{\sigma_u(u_0, v_0), \sigma_v(u_0, v_0)\}$. Shrinking U if necessary, we have $\tilde{\sigma} = \sigma \circ (\sigma^{-1} \circ \tilde{\sigma}) = \sigma \circ \Phi$ for $\Phi := \sigma^{-1} \circ \tilde{\sigma} : \tilde{U} \rightarrow U$, where in coordinates we have $\Phi(\tilde{u}, \tilde{v}) = (u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v}))$. Φ is smooth since σ and $\tilde{\sigma}$ are diffeomorphisms. Thus by the chain rule,

$$\tilde{\sigma}_{\tilde{u}}(\tilde{u}, \tilde{v}) = \sigma_u(u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v})) \frac{\partial u}{\partial \tilde{u}}(\tilde{u}, \tilde{v}) + \sigma_v(u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v})) \frac{\partial v}{\partial \tilde{u}}(\tilde{u}, \tilde{v}) \in \text{Span}\{\sigma_u(u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v})), \sigma_v(u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v}))\}$$

where the multiplication is the appropriate chain rule one, and similarly,

$$\tilde{\sigma}_{\tilde{v}}(\tilde{u}, \tilde{v}) = \sigma_u(u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v})) \frac{\partial u}{\partial \tilde{v}}(\tilde{u}, \tilde{v}) + \sigma_v(u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v})) \frac{\partial v}{\partial \tilde{v}}(\tilde{u}, \tilde{v}) \in \text{Span}\{\sigma_u(u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v})), \sigma_v(u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v}))\}$$

and so $\text{Span}\{\sigma_u(u_0, v_0), \sigma_v(u_0, v_0)\} \supseteq \text{Span}\{\tilde{\sigma}_u(\tilde{u}_0, \tilde{v}_0), \tilde{\sigma}_v(\tilde{u}_0, \tilde{v}_0)\}$, and so for dimensionality considerations these are equal (one is a subset and they have the same dimension).

(3) We have

$$\begin{aligned} \tilde{\sigma}_{\tilde{u}} \times \tilde{\sigma}_{\tilde{v}} &= \left(\sigma_u \frac{\partial u}{\partial \tilde{u}} + \sigma_v \frac{\partial v}{\partial \tilde{u}} \right) \times \left(\sigma_u \frac{\partial u}{\partial \tilde{v}} + \sigma_v \frac{\partial v}{\partial \tilde{v}} \right) \\ &= \left(\frac{\partial u}{\partial \tilde{u}} \frac{\partial v}{\partial \tilde{v}} \right) \sigma_u \times \sigma_v + \left(\frac{\partial v}{\partial \tilde{u}} \frac{\partial u}{\partial \tilde{v}} \right) \sigma_v \times \sigma_u = \left(\frac{\partial u}{\partial \tilde{u}} \frac{\partial v}{\partial \tilde{v}} - \frac{\partial v}{\partial \tilde{u}} \frac{\partial u}{\partial \tilde{v}} \right) \sigma_u \times \sigma_v = \det(D\Phi) \sigma_u \times \sigma_v \end{aligned}$$

so normalizing it, we see the cross products are the same, up to the sign of the transition function Φ 's Jacobian (this fact is itself useful, a shorter proof can just be because the spans are equal so the plane has exactly two normal unit vectors, one for each sign). ■

Chapter 3: Submanifolds of $\mathbb{R}^n$

Definition

A k -dimensional (topological) submanifold of $\mathbb{R}^n$ is a nonempty subset $M \subseteq \mathbb{R}^n$ such that for all $p \in M$, there exists an open set $U \subseteq \mathbb{R}^k$ and an open set $W \subseteq \mathbb{R}^n$ with $p \in W$ and a homeomorphism $\sigma : U \rightarrow M \cap W$. The map σ is called a coordinate chart. An atlas for M is a collection of charts $\{\sigma_a : U_a \rightarrow M \cap W_a\}_{a \in \mathcal{A}}$ of charts such that $\bigcup_{a \in \mathcal{A}} (M \cap W_a) = M$. We call k the dimension of M , and write $\dim(M) = k$. It is a fact that k is unique (by algebraic topology).

Example

$M = \mathbb{R}^n$ is an n -dimensional submanifold of $\mathbb{R}^n$ (Take $U = \mathbb{R}^n = W$, σ to be the identity).

Example

More generally, any open set in $\mathbb{R}^n$ is an n -dimensional submanifold of $\mathbb{R}^n$, again with the identity map

(restricted) as the chart.

Remark

Although M is not assumed to be connected, the k appearing in the definition is fixed across the connected components of M .

Example

A hyperplane Π of the form $a_1x_1 + \cdots + a_nx_n = b$ with $(a_1, \dots, a_n) \neq 0$ a normal vector Π is an $n - 1$ dimensional submanifold of $\mathbb{R}^n$. Indeed, without loss of generality suppose $a_1 \neq 0$. Then $x_1 = \frac{1}{a_1}(b - a_2x_2 - \cdots - a_nx_n)$ for all $(x_1, \dots, x_n) \in \Pi$, so it is graph of $f : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$ given by $f(x_2, \dots, x_n) = \frac{1}{a_1}(b - a_2x_2 - \cdots - a_nx_n)$. Then as in the $\mathbb{R}^3$ case, the chart $\sigma : U = \mathbb{R}^{n-1} \rightarrow \Pi \subseteq \mathbb{R}^n$ by $\sigma(x_2, \dots, x_n) = (f(x_2, \dots, x_n), x_2, \dots, x_n)$ works.

Example

More generally, the graph of any continuous function $f : U \subseteq \mathbb{R}^k \rightarrow \mathbb{R}^{n-k}$ (U open), $G = \{(x, y) \in \mathbb{R}^k \times \mathbb{R}^{n-k} : y = f(x)\} \subseteq \mathbb{R}^n$ is a k -submanifold of $\mathbb{R}^n$, with $x \mapsto (x, f(x))$ as the global chart.

Example

The twisted cubic, $C = \{(x, y, z) \in \mathbb{R}^3 : y = x^2, z = x^3\}$ is the graph of the continuous $f : \mathbb{R} \rightarrow \mathbb{R}^2$ given by $f(x) = (x^2, x^3)$, and so is a 1-submanifold of $\mathbb{R}^{1+2} = \mathbb{R}^3$.

Definition

A hypersurface is an $n - 1$ dimensional submanifold of $\mathbb{R}^n$. For example, surfaces in $\mathbb{R}^3$, hyperplanes, and the $n - 1$ -sphere, $S^{n-1} = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_1^2 + \cdots + x_n^2 = 1\} \subseteq \mathbb{R}^n$.

Definition

A smooth k -dimensional submanifold of $\mathbb{R}^n$ is a nonempty set $M \subseteq \mathbb{R}^n$ such that for all $p \in M$, there exists open sets $U \subseteq \mathbb{R}^k$, $W \subseteq \mathbb{R}^n$ with $p \in W$, and a homeomorphism $\sigma : U \rightarrow M \cap W$ such that σ is smooth, and σ 's partial derivatives are linearly independent for every point in $u \in U$, or equivalently, $D\sigma(u)$ has rank k (maximal rank) at every $u \in U$. Such a σ is called a smooth chart for M , and an atlas of M made of smooth charts is called a smooth atlas.

Example

$\mathbb{R}^n$ is a smooth n -dimensional submanifold of $\mathbb{R}^n$, where σ is the (smooth) identity map, $U = \mathbb{R}^n = W$, and $D\sigma(u) = I_{n \times n}$ for all $u \in U$.

Example

More generally, any $U \subseteq \mathbb{R}^n$ open is a smooth n -submanifold of $\mathbb{R}^n$, where the (global) chart is the restriction of the (smooth) identity map.

Example

Graphs of smooth functions $f : U \subseteq \mathbb{R}^k \rightarrow \mathbb{R}^{n-k}$ (U open), $G = \{(x, y) \in \mathbb{R}^k \times \mathbb{R}^{n-k} : y = f(x)\} \subseteq \mathbb{R}^n$ is a k -submanifold of $\mathbb{R}^n$. Its (global) chart is $\sigma : \mathbb{R}^k \rightarrow G \subseteq \mathbb{R}^n$ given by $\sigma(x) = (x, f(x))$ for $x \in \mathbb{R}^k$, which is smooth since f is, where its derivative is $D\sigma(x) = \begin{bmatrix} I_{k \times k} \\ Df(x) \end{bmatrix}$ (which is an $n \times k$ matrix), which has rank k because of the identity block. Thus, $T_pM = \text{Span} \left\{ \begin{bmatrix} e_i \\ \frac{\partial f}{\partial x_i} \end{bmatrix} : i = 1, \dots, k \right\}$.

Definition

Let M be a smooth submanifold of $\mathbb{R}^n$ and $p \in M$. Choose a smooth chart $\sigma : U \rightarrow M \cap W$, with coordinates $u_1, \dots, u_k$, for $p \in M \cap W$. If $a = \sigma^{-1}(p)$, we set $T_pM = \text{Span} \left\{ \frac{\partial \sigma}{\partial u_1}(a_0), \dots, \frac{\partial \sigma}{\partial u_k}(a_0) \right\}$, the tangent space to M at p . The proof we saw for the surfaces case transfers over easily to show that T_pM is independent of the choice of chart σ , and $\dim(T_pM) = \dim M$ because $\left\{ \frac{\partial \sigma}{\partial u_1}(a_0), \dots, \frac{\partial \sigma}{\partial u_k}(a_0) \right\}$ is linearly

independent by definition of σ . Moreover, as shown previously for the case of surfaces, the tangent space $T_p M$ can be considered as the set of velocity vectors $\{\gamma'(0) : \gamma : (-\varepsilon, \varepsilon) \rightarrow M \text{ with } \gamma(0) = p, \gamma \text{ smooth}\}$, and the same proof works.

As we showed in the case of smooth surfaces,

Definition

An equivalent definition of a smooth k -dimensional submanifold $M \subseteq \mathbb{R}^n$ is, for all $p \in M$, there exists a smooth function $F : W \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$ (W open), such that $p \in W$, and $M \cap W$ is the zero set of F and DF has rank $(n - k)$ identically on $M \cap W$.

To see one direction of this equivalence, let $x_0 \in M \cap W$, where then $DF(x_0)$ has rank $n - k$ in a neighbourhood of x_0 in $M \cap W$. Then by the implicit function theorem, we can express $n - k$ of the variables as smooth functions of the remaining k variables. That is, if $(x_1, \dots, x_n)$ are the coordinates in $\mathbb{R}^n$, then around x_0 in $M \cap W$, we can (permuting the coordinates if necessary) write $x_{k+1} = f_1(x_1, \dots, x_k)$, $x_{k+2} = f_2(x_1, \dots, x_k)$, $\dots$, $x_n = f_{n-k}(x_1, \dots, x_k)$. Then if we set $f : U \subseteq \mathbb{R}^k \rightarrow \mathbb{R}^{n-k}$ by $u_1, \dots, u_k \mapsto (f_1(u_1, \dots, u_k), \dots, f_{n-k}(u_1, \dots, u_k))$, and its graph is a neighbourhood of x_0 in $M \cap W$, so we see M is a smooth k -submanifold by the original definition.

All-in-all:

Theorem

Let $\emptyset \neq M \subseteq \mathbb{R}^n$. The following are equivalent:

- (1) M is a smooth k -dimensional submanifold (in terms of our original definition with smooth charts).
- (2) M is locally the graph of smooth functions $f : U \subseteq \mathbb{R}^k \rightarrow \mathbb{R}^{n-k}$ (U open, can vary for choice of f).
- (3) M is locally the zero set of smooth functions $F : W \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$ (W open, can vary for choice of F) where DF has rank $n - k$ in the neighbourhood we are considering.

Example

Consider the unit $(n - 1)$ -sphere $S^{n-1} \subseteq \mathbb{R}^n$, $S^{n-1} = \{(x_1, \dots, x_n) : x_1^2 + \dots + x_n^2 = 1\} = \{(x_1, \dots, x_n) : F(x_1, \dots, x_n) = 0\}$ for $F(x_1, \dots, x_n) = x_1^2 + \dots + x_n^2 - 1$. Then take $k = n - 1$, and $DF = \nabla F = (2x_1, \dots, 2x_n) \neq 0$ on S^{n-1} , so we see S^{n-1} is a smooth n -dimensional submanifold of $\mathbb{R}^n$.

Example

Consider $\text{SL}(2, \mathbb{R})$, the group of 2×2 matrices with determinant 1 (special linear group). Identifying $M_{2 \times 2}(\mathbb{R}) \cong \mathbb{R}^4$ by $\begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} = (a_1, a_2, a_3, a_4) \in \mathbb{R}^4$. We claim $\text{SL}(2, \mathbb{R})$ is a smooth 3-manifold of $\mathbb{R}^4$.

Then for $A \in \mathbb{R}^4$, $\det A = a_1 a_4 - a_2 a_3$, and $\text{SL}(2, \mathbb{R}) = \{(a_1, \dots, a_4) \in \mathbb{R}^4 : a_1 a_4 - a_2 a_3 = 1\} = \{(a_1, \dots, a_4) \in \mathbb{R}^4 : a_1 a_4 - a_2 a_3 - 1 = 0\}$. Thus define $F : \mathbb{R}^4 \rightarrow \mathbb{R}$ by $(a_1, \dots, a_4) \mapsto a_1 a_4 - a_2 a_3 - 1$, and $DF = \nabla F = (a_4, -a_3, -a_2, a_1)$ which equals 0 if and only if $a_1 = a_2 = a_3 = a_4 = 0$, which does not correspond to a matrix of determinant 1 (it corresponds to the zero matrix), so DF has maximal rank on $\text{SL}(2, \mathbb{R})$ and so $\text{SL}(2, \mathbb{R})$ is a smooth $(4 - 1) = 3$ dimensional submanifold of $\mathbb{R}^4$.

Another important interpretation of the tangent space:

We will show on the assignments that if M is globally the zero set of $F : W \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$, with DF maximal rank on M , then $T_p M = \ker DF(p)$. In particular, if $k = n - 1$ so that M is a hypersurface and DF is the gradient vector. Then $\ker DF(p)$ is the set of vectors $v \in \mathbb{R}^n$ such that $DF(p)v = 0$, but $DF(p)v = \nabla F(p)v$, which is the usual dot product (v is a column, DF is a row), so this shows $T_p M$ is the set of vectors in $\mathbb{R}^n$ orthogonal to $\nabla F(p)$ ($\nabla F(p)$ never vanishes on M by definition).

Chapter 4: Surfaces

For the entire chapter, S denotes a *smooth* surface in $\mathbb{R}^3$.

Let $\sigma : U \subseteq \mathbb{R}^2 \rightarrow S \cap W \subseteq \mathbb{R}^3$ (U, W open in $\mathbb{R}^2$ and $\mathbb{R}^3$ respectively) be a smooth regular coordinate chart containing $p \in S$, and recall the standard unit normal of S at $p = \sigma(u_0, v_0)$ is

$$N_\sigma(u_0, v_0) = \frac{\sigma_u(u_0, v_0) \times \sigma_v(u_0, v_0)}{\|\sigma_u(u_0, v_0) \times \sigma_v(u_0, v_0)\|}$$

Example

Let $S = \{(x, y, z) \in \mathbb{R}^3 : z = x^2 + y^2\} \subseteq \mathbb{R}^3$ be a paraboloid, and parameterize this with the chart $\sigma(u, v) = (u, v, u^2 + v^2)$ for $(u, v) \in \mathbb{R}^2$. Then $\sigma_u(u, v) = (1, 0, 2u)$, $\sigma_v(u, v) = (0, 1, 2v)$, and let $p = \sigma(u_0, v_0) = (u_0, v_0, u_0^2 + v_0^2)$. Then

$$N_\sigma(u_0, v_0) = \frac{\sigma_u(u_0, v_0) \times \sigma_v(u_0, v_0)}{\|\sigma_u(u_0, v_0) \times \sigma_v(u_0, v_0)\|} = \frac{1}{\sqrt{4u_0^2 + 4v_0^2 + 1}}(-2u_0, -2v_0, 1)$$

This should make intuitive sense if we consider the paraboloid and p being the origin ($u_0 = v_0 = 0$). Then the tangent plane should be the xy -plane, so its normal vector should be (up to a sign) in the direction of the z -axis, which is exactly what we got above.

We have seen that N_σ is independent of choice of chart, up to a sign. How can we control the sign? Suppose that $\sigma : U \subseteq \mathbb{R}^2 \rightarrow S \cap W \subseteq \mathbb{R}^3$ is a smooth regular chart with coordinates $(u, v) \mapsto (x(u, v), y(u, v), z(u, v))$, and let $\tilde{\sigma} : \tilde{U} \subseteq \mathbb{R}^2 \rightarrow S \cap \tilde{W} \subseteq \mathbb{R}^3$ be another smooth regular chart with coordinates $(\tilde{u}, \tilde{v}) \mapsto (x(\tilde{u}, \tilde{v}), y(\tilde{u}, \tilde{v}), z(\tilde{u}, \tilde{v}))$, with $W \cap \tilde{W}$ nonempty. Set $\Phi = \sigma^{-1} \circ \tilde{\sigma} : \tilde{\sigma}^{-1}(S \cap W \cap \tilde{W}) \rightarrow \sigma^{-1}(S \cap W \cap \tilde{W})$, which maps points $(\tilde{u}, \tilde{v}) \mapsto (u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v}))$. Then $\tilde{\sigma} = \sigma \circ \Phi$ on the appropriate overlap.

Example

As an example, consider parameterizing the xy -plane S in $\mathbb{R}^3$, we have two charts $\sigma : \mathbb{R} \rightarrow S$ by $(u, v) \mapsto (u, v, 0)$ and polar coordinates, $\tilde{\sigma} : (0, \infty) \times (0, 2\pi) \rightarrow S \setminus \{\text{positive } x\text{-axis}\}$, by $(\tilde{u}, \tilde{v}) \mapsto (\tilde{u} \cos \tilde{v}, \tilde{u} \sin \tilde{v}, 0)$. Then

$$\Phi(\tilde{u}, \tilde{v}) = \sigma^{-1} \circ \tilde{\sigma}(\tilde{u}, \tilde{v}) = (\tilde{u} \cos \tilde{v}, \tilde{u} \sin \tilde{v}) = (u, v)$$

Returning to the above, $\tilde{\sigma} = \sigma \circ \Phi$, by the chain rule, $D\tilde{\sigma} = D\sigma D\Phi$. Now, Φ is a map from an open subset of $\mathbb{R}^2$ to an open subset of $\mathbb{R}^2$, and is a diffeomorphism, so $D\Phi$ is a 2×2 invertible matrix, so $D\tilde{\sigma}$ and $D\sigma$ differ by an invertible matrix, and our calculation before reading week (in proving the normal vector was independent of choice of chart),

$$N_{\tilde{\sigma}} = \frac{\tilde{\sigma}_{\tilde{u}} \times \tilde{\sigma}_{\tilde{v}}}{\|\tilde{\sigma}_{\tilde{u}} \times \tilde{\sigma}_{\tilde{v}}\|} = \frac{\det D\Phi}{|\det D\Phi|} \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|} = \frac{\det D\Phi}{|\det D\Phi|} N_\sigma = \text{sgn}(\det D\Phi) N_\sigma$$

So the signs differ by exactly the sign of the determinant of the derivative of the transition function.

Chapter 4.1: First Fundamental Form

Definition

Let S be a smooth surface in $\mathbb{R}^3$, and let $p \in S$. We know $T_p S$ is a 2-dimensional subspace of $\mathbb{R}^3$. We know for any $x, y \in \mathbb{R}^3$, we have $x \cdot y = \sum_{i=1}^3 x_i y_i$ as the Euclidean inner product. We can

restrict this to $x, y \in T_p S$. For some chart σ of S with $p = \sigma(u_0, v_0)$ and coordinates (u, v) , write $x = a\sigma_u(u_0, v_0) + b(\sigma_v(u_0, v_0))$, $y = c\sigma_u(u_0, v_0) + d\sigma_v(u_0, v_0)$. Then

$$x \cdot y = (a\sigma_u + b\sigma_v) \cdot (c\sigma_u + d\sigma_v) = ac\sigma_u \cdot \sigma_u + ad\sigma_u \cdot \sigma_v + bc\sigma_v \cdot \sigma_u + bd\sigma_v \cdot \sigma_v$$

symmetry of $\cdot$

$$\stackrel{\text{symmetry of } \cdot}{=} ac \underbrace{\sigma_u \cdot \sigma_u}_E + (ad + bc) \underbrace{\sigma_u \cdot \sigma_v}_F + bd \underbrace{\sigma_v \cdot \sigma_v}_G = acE + (ad + bc)F + bdG = \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} E & F \\ F & G \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix}$$

where $E = \sigma_u \cdot \sigma_u$, $F = \sigma_u \cdot \sigma_v$, $G = \sigma_v \cdot \sigma_v$. The matrix $\mathcal{F}_I := \begin{bmatrix} E & F \\ F & G \end{bmatrix} = \begin{bmatrix} \sigma_u \cdot \sigma_u & \sigma_u \cdot \sigma_v \\ \sigma_u \cdot \sigma_v & \sigma_v \cdot \sigma_v \end{bmatrix}$ is called the First Fundamental Form of σ , and is a symmetric 2×2 matrix. This is the matrix representation of the restriction of the dot product $\cdot$ in $\mathbb{R}^3$ to $T_p S$ in terms of the basis $\mathcal{B} := \{\sigma_u(u_0, v_0), \sigma_v(u_0, v_0)\}$ of $T_p S$. Indeed, if $x = a\sigma_u + b\sigma_v$, $y = c\sigma_u + d\sigma_v \in T_p S$, then $[x]_{\mathcal{B}} = (a, b)^T$, $[y]_{\mathcal{B}} = (c, d)^T$, and $x \cdot y = [x]_{\mathcal{B}}^T \mathcal{F}_I [y]_{\mathcal{B}}$.

Example

Consider a plane Π in $\mathbb{R}^3$, determined by a point $p_0 \in \mathbb{R}^3$ with direction vectors $w_1, w_2 \in \mathbb{R}^3$. A smooth regular (global) chart for Π is $\sigma(u, v) = p_0 + uw_1 + vw_2$ for $(u, v) \in \mathbb{R}^2$. Let $p \in \Pi$, with $\sigma(u_0, v_0) = p$. Then $\sigma_u(u_0, v_0) = w_1$, $\sigma_v(u_0, v_0) = w_2$, and $T_p S = \text{Span}\{\sigma_u(u_0, v_0), \sigma_v(u_0, v_0)\} = \text{Span}\{w_1, w_2\}$ and $\mathcal{F}_I = \begin{bmatrix} w_1 \cdot w_1 & w_1 \cdot w_2 \\ w_1 \cdot w_2 & w_2 \cdot w_2 \end{bmatrix}$. In particular, if we choose direction vectors w_1, w_2 that are orthogonal and of norm 1, we get $\mathcal{F}_I = I_{2 \times 2}$.

Example

Consider a cylinder described by the equation $x^2 + y^2 = 1$. This has smooth regular chart $\sigma(u, v) = (\cos u, \sin u, v)$ with $v \in \mathbb{R}$ and $u \in (0, 2\pi)$ or $u \in (-\pi, \pi)$ (together, this chart with these two domains forms an atlas). Then $\sigma_u(u, v) = (-\sin u, \cos u, 0)$, and $\sigma_v(u, v) = (0, 0, 1)$. Then $\sigma_u \cdot \sigma_u = 1$, $\sigma_u \cdot \sigma_v = 0$, $\sigma_v \cdot \sigma_v = 1$, so $\mathcal{F}_I = I_{2 \times 2}$ for this chart. This shows the First Fundamental Form is not enough to define curvature of a surface, it is the same as in our plane, but cylinders are curved but planes aren't.

Example

Consider the sphere $x^2 + y^2 + z^2 = a^2$ with $a > 0$. This has a smooth regular chart $\sigma(\theta, \phi) = (a \sin \phi \cos \theta, a \sin \phi \sin \theta, a \cos \phi)$, with $\theta \in (0, 2\pi)$, $\phi \in (0, \pi)$, where $\sigma_\theta(\theta, \phi) = (-a \sin \phi \sin \theta, a \sin \phi \cos \theta, 0)$, $\sigma_\phi(\theta, \phi) = (a \cos \phi \cos \theta, a \cos \phi \sin \theta, -a \sin \phi)$. Then $\sigma_\theta \cdot \sigma_\theta = a^2 \sin^2 \phi (\sin^2 \theta + \cos^2 \theta) = a^2 \sin^2 \phi$, and $\sigma_\theta \cdot \sigma_\phi = 0$, $\sigma_\phi \cdot \sigma_\phi = a^2$, so $\mathcal{F}_I = \begin{bmatrix} a^2 \sin^2 \phi & 0 \\ 0 & a^2 \end{bmatrix}$.

Suppose that $\tilde{\sigma}$ is another coordinate chart of S and that $\tilde{\sigma} = \sigma \circ \Phi$ for some transition function Φ (a diffeomorphism from open subset of $\mathbb{R}^2$ to open subset of $\mathbb{R}^2$). Then, if $\tilde{\mathcal{F}}_I$ is the first fundamental form of $\tilde{\sigma}$, we have that $\tilde{\mathcal{F}}_I = (D\Phi)^T \mathcal{F}_I D\Phi$.

To see this, notice that for $x, y \in T_p S$, then if we write $x = (a, b)_{\mathcal{B}}$, $y = (c, d)_{\mathcal{B}}$ with respect to the basis $\mathcal{B}$, where $\mathcal{B} = \{\sigma_u, \sigma_v\}$ for a smooth regular chart σ with coordinates u, v , then $x = D\sigma(a, b)^T$, $y = D\sigma(c, d)^T$, so

$$x \cdot y = x^T y = (D\sigma(a, b)^T)^T (D\sigma(c, d)^T) = (a, b) D\sigma^T D\sigma (c, d)^T$$

so $\mathcal{F}_I = D\sigma^T D\sigma$ (this is independently useful). Then, for some other chart $\tilde{\sigma}$ with $\tilde{\sigma} = \sigma \circ \Phi$ for a diffeomorphism Φ , the first fundamental form in that chart is $\tilde{\mathcal{F}}_I = (D\tilde{\sigma})^T D\tilde{\sigma}$, so by the chain rule,

$$\tilde{\mathcal{F}}_I = (D\tilde{\sigma})^T D\tilde{\sigma} = (D(\sigma \circ \Phi))^T D(\sigma \circ \Phi) = (D\sigma D\Phi)^T D\sigma D\Phi = D\Phi^T D\sigma^T D\sigma D\Phi = D\Phi^T \mathcal{F}_I D\Phi$$

Let us consider a smooth curve on S , say we have $\gamma : (\alpha, \beta) \rightarrow S$. Without loss of generality, say that the image of γ is contained in a single coordinate patch, that is, $\text{im } \gamma \subseteq S \cap W$ for some open set $W \subseteq \mathbb{R}^3$, with $\sigma : U \subseteq \mathbb{R}^2 \rightarrow S \cap W$ a smooth regular chart. Then, we get a curve $\bar{\gamma} := \sigma^{-1} \circ \gamma : (\alpha, \beta) \rightarrow U$, which is smooth since σ^{-1} and γ are smooth. Write the component functions of $\bar{\gamma}$ as $u(t), v(t)$. Then we

have $\gamma(t) = (\sigma \circ \bar{\gamma})(t) = \sigma(u(t), v(t))$ for all $t \in (\alpha, \beta)$. Therefore, any smooth curve on $S \cap W$ is of the form $\gamma(t) = \sigma(u(t), v(t))$ for some smooth scalar functions $u, v : (\alpha, \beta) \rightarrow \mathbb{R}$. In particular, by the chain rule, $\gamma'(t) = D\sigma\bar{\gamma}'(t)$, and

$$\begin{aligned} \|\gamma'(t)\|^2 &= \gamma'(t) \cdot \gamma'(t) = (D\sigma\bar{\gamma}'(t))^T (D\sigma\bar{\gamma}'(t)) = \bar{\gamma}'(t)^T D\sigma^T D\sigma\bar{\gamma}'(t) = \bar{\gamma}'(t)^T \mathcal{F}_I \bar{\gamma}'(t) \\ &= \|\bar{\gamma}'(t)\|_{\mathcal{F}_I}^2 \end{aligned}$$

where the subscript $\mathcal{F}_I$ denotes the norm with respect to the inner product induced by $\mathcal{F}_I$ on the tangent space $T_{\gamma(t)}S$. Using this, we can calculate the arclength of γ :

$$\text{arclength} = \int_{\alpha}^{\beta} \|\gamma'(t)\| dt = \int_{\alpha}^{\beta} \|\bar{\gamma}'(t)\|_{\mathcal{F}_I} dt = \text{length of } \bar{\gamma} \text{ in } \mathbb{R}^2 \text{ w.r.t. } \|\cdot\|_{\mathcal{F}_I}$$

Thus, we can ask about maps between surfaces that preserve this notion of length, which are called isometries. What do we mean by a map between surfaces?

Definition

Let S and S' be two smooth surfaces in $\mathbb{R}^3$. A map $F : S \subseteq \mathbb{R}^3 \rightarrow S' \subseteq \mathbb{R}^3$ with components $(x, y, z) \mapsto (F_1(x, y, z), F_2(x, y, z), F_3(x, y, z))$ is smooth if the component functions $F_1, F_2, F_3 : W \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}$ are smooth scalar functions in an open set W containing S .

Example

The identity map $\text{Id}_S : S \rightarrow S$ by $(x, y, z) \mapsto (x, y, z)$ is smooth (the component functions are the projections $(x, y, z) \mapsto x, (x, y, z) \mapsto y, (x, y, z) \mapsto z$ which are smooth on $\mathbb{R}^3$ which is open and contains S).

Example

Let $S = \{(x, y, z) \in \mathbb{R}^3 : z = x^2 + y^2\}, S' = \{(x, y, 0) \in \mathbb{R}^3\}$ (the xy -plane). $F : S \rightarrow S'$ by $(x, y, z) = (x, y, x^2 + y^2) \mapsto (x^2y - 1, x - y - z, 0)$ is smooth (any scalar functions smooth in x, y, z work for the first two component functions).

Example

Let $S = \{(x, y, 0) \in \mathbb{R}^3 : (x, y) \in (0, 2\pi) \times \mathbb{R}\}$ be an open strip of the xy -plane. Define $S' = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\} \setminus \{(x, y, z) \in \mathbb{R}^3 : x = 1, y = 0\}$ (the cylinder a vertical line). Consider the map $F : S \rightarrow S'$ by $(x, y, 0) \mapsto (\cos x, \sin x, y)$, which is smooth.

Definition

A map between smooth surfaces S and S' , $F : S \rightarrow S'$ is a diffeomorphism if it is a smooth bijection whose inverse, $F^{-1} : S' \rightarrow S$ is smooth.

Example

The identity map $\text{Id} : S \rightarrow S$ is a diffeomorphism.

Example

Consider the antipodal map $F : S^2 \rightarrow S^2$ given by $(x, y, z) \mapsto (-x, -y, -z)$ where S^2 is the unit sphere. This is a smooth map from S^2 to S^2 , and is a bijection whose inverse is itself, and thus has a smooth inverse, and thus F is a diffeomorphism.

Example

For any $a > 0$, a map $F_a : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $(x, y, z) \mapsto (ax, ay, az)$ is called a dilation if $a > 1$ and a contraction if $a < 1$, which is a smooth bijection with smooth inverse $(x, y, z) \mapsto (x/a, y/a, z/a)$, and thus is a diffeomorphism of a surface S provided its image is a smooth surface, for example, it is a diffeomorphism from S^2 to S_a^2 (the sphere of radius a in $\mathbb{R}^3$).

Proposition

Let $F : S \rightarrow S'$ be a diffeomorphism of smooth surfaces S and S' . Then, for any smooth coordinate

chart $\sigma : U \subseteq \mathbb{R}^2 \rightarrow S \cap W \subseteq \mathbb{R}^3$ of S , the composition $\sigma' := F \circ \sigma : U \rightarrow S' \cap F(W)$ is a smooth chart for S' .

Proof:

Note F is a diffeomorphism and thus a homeomorphism, and so σ' is a homeomorphism, which shows $S' \cap F(W)$ is indeed open in S' (i.e. for anyone reading my notes $F(W)$ is open in $\mathbb{R}^3$). Since σ and F are both smooth, we also get that σ' is smooth. Thus, it remains to check it is regular. Then $D\sigma' = D(F \circ \sigma) = DF D\sigma$ by the chain rule, but $D\sigma$ has rank 2 by definition of σ , and DF is invertible since F is a diffeomorphism ($F \circ F^{-1} = \text{Id}$, implying by the chain rule $D(F^{-1}) = (DF)^{-1}$), and so by linear algebra, $D\sigma' = DF D\sigma$ has rank 2, so σ' is a smooth coordinate chart for S' . ■

Definition

Let S, S' be two smooth surfaces, and let $F : S \rightarrow S'$ be a diffeomorphism. Let $\sigma : U \subseteq \mathbb{R}^2 \rightarrow S \cap W \subseteq \mathbb{R}^3$ be any smooth chart on S , and set $\sigma' := F \circ \sigma$. Let $\mathcal{F}_I, \mathcal{F}'_I$ be the first fundamental forms of σ and σ' respectively.

- (i) If $\mathcal{F}_I = \mathcal{F}'_I$, then F is called an isometry.
- (ii) If $\mathcal{F}_I = f \mathcal{F}'_I$ for some positive scalar function $f : U \rightarrow \mathbb{R}^+$, then F is called a conformal map. By taking $f \equiv 1$, we see every isometry is conformal.

Remark

We will show that isometries preserve lengths, that is, if $\gamma : (\alpha, \beta) \rightarrow S$ is a smooth curve, then $\delta := F \circ \gamma : (\alpha, \beta) \rightarrow S'$, then γ and δ have the same arc length if F is an isometry, and if F is conformal, then F preserves angles.

Example

Not all conformal maps are isometries. The dilation $F_a : S^2 \rightarrow S_a^2$ for $a > 1$ we saw above is conformal but not an isometry: Given any chart $\sigma : U \rightarrow S^2 \cap W$ for S^2 , we have $\sigma' = a\sigma$. Then $\mathcal{F}_I = (D\sigma^T)D\sigma$, but $\mathcal{F}'_I = (D\sigma'^T)D\sigma' = (aD\sigma)^T a D\sigma = a^2 D\sigma^T D\sigma = a^2 \mathcal{F}_I \neq \mathcal{F}_I$ since $a > 1$. Thus F_a is not an isometry, but is a conformal map since $(x, y) \mapsto a^2$ is a positive scalar function.

Example

Consider the strip of the xy -plane, $S = \{(x, y, 0) : (x, y) \in (0, 2\pi) \times \mathbb{R}\} \subseteq \mathbb{R}^3$ and the cylinder minus a vertical line, $S' = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\} \setminus \{(x, y, z) \in \mathbb{R}^3 : x = 1, y = 0\}$. Consider the diffeomorphism $F : S \rightarrow S'$ given by $(x, y, 0) \mapsto (\cos x, \sin x, y)$. We claim this is an isometry. We must check that if $\sigma : U \rightarrow S \cap W$ is any chart of S and $\tilde{\sigma} = F \circ \sigma$, then $\mathcal{F}_I = \tilde{\mathcal{F}}_I$. Because σ is a smooth chart for S , we have $\sigma(u, v) = (x(u, v), y(u, v), 0)$ for all $(u, v) \in U$, for some smooth functions $x, y : U \rightarrow \mathbb{R}$. Then $\tilde{\sigma}(u, v) = (\cos(x(u, v)), \sin(x(u, v)), y(u, v))$. Then $\sigma_u = (x_u, y_u, 0)$, $\sigma_v = (x_v, y_v, 0)$, so $\mathcal{F}_I = \begin{bmatrix} x_u^2 + y_u^2 & x_u x_v + y_u y_v \\ x_u x_v + y_u y_v & x_v^2 + y_v^2 \end{bmatrix}$. Then $\tilde{\sigma}_u = (-x_u \sin(x), x_u \cos(x), y_u)$, and $\tilde{\sigma}_v = (-x_v \sin(x), x_v \cos(x), y_v)$, so $\tilde{\mathcal{F}}_I = \begin{bmatrix} x_u^2 + y_u^2 & x_u x_v + y_u y_v \\ x_u x_v + y_u y_v & x_v^2 + y_v^2 \end{bmatrix} = \mathcal{F}_I$, so F is an isometry by definition.

Proposition

- (1) Isometries preserve distances
- (2) Conformal maps preserve angles.

Proof:

Let $F : S \rightarrow S'$ be an isometry. Let $\sigma : U \rightarrow S \cap W$ be a smooth chart and set $\tilde{\sigma} = F \circ \sigma$. Let $\gamma : (\alpha, \beta) \rightarrow S \cap W$ be a smooth curve and set $\tilde{\gamma} = F \circ \gamma$. Then the length of γ on $[\alpha_0, \beta_0] \subseteq (\alpha, \beta)$ is $L_\gamma := \int_{\alpha_0}^{\beta_0} \|\gamma'(t)\| dt = \int_{\alpha_0}^{\beta_0} \|\tilde{\gamma}'(t)\|_{\mathcal{F}_I} dt$, where $\tilde{\gamma} = \sigma^{-1} \circ \gamma$, as shown previously. Similarly, $\tilde{\sigma}^{-1} \circ \tilde{\gamma} = \sigma^{-1} \circ F^{-1} \circ F \circ \gamma = \sigma^{-1} \circ \gamma = \tilde{\gamma}$, so $L_{\tilde{\gamma}} := \int_{\alpha_0}^{\beta_0} \|\tilde{\gamma}'(t)\|_{\tilde{\mathcal{F}}_I} dt$, so since $\tilde{\mathcal{F}}_I = \mathcal{F}_I$, these integrals are equal, which proves (1).

Proving (2), suppose that F is a conformal map so that $\mathcal{F}_I = f \tilde{\mathcal{F}}_I$ some positive scalar function $f : U \rightarrow$

$\mathbb{R}$. Let $\gamma_1, \gamma_2 : (\alpha, \beta) \rightarrow S \cap W$ be two smooth curves on $S \cap W$. Set $\tilde{\gamma}_1 = F \circ \gamma_1, \tilde{\gamma}_2 = F \circ \gamma_2$, which are smooth curves on some patch $S' \cap W'$ of S' . Suppose that γ_1, γ_2 intersect at $\gamma_1(t_0) = \gamma_2(t_0) = p \in S \cap W$, so $\tilde{\gamma}_1, \tilde{\gamma}_2$ intersect at $F(p) = \tilde{\gamma}_1(t_0) = \tilde{\gamma}_2(t_0)$. Let θ be the angle of the intersection between γ_1 and γ_2 at p , and so

$$\cos \theta = \frac{\gamma'_1(t_0) \cdot \gamma'_2(t_0)}{\|\gamma'_1(t_0)\| \|\gamma'_2(t_0)\|}$$

with $\gamma'_1(t_0) \cdot \gamma'_2(t_0) = (\bar{\gamma}'_1(t_0))^T \mathcal{F}_I \bar{\gamma}'_2(t_0)$ (using the notation above), and $\|\gamma'_1(t_0)\| = \|\bar{\gamma}'_1(t_0)\|_{\mathcal{F}_I}$ and $\|\gamma'_2(t_0)\| = \|\bar{\gamma}'_2(t_0)\|_{\mathcal{F}_I}$.

Thus,

$$\tilde{\gamma}'_1(t_0) \cdot \tilde{\gamma}'_2(t_0) = (\bar{\gamma}'_1(t_0))^T \tilde{\mathcal{F}}_I \bar{\gamma}'_2(t_0) = (\bar{\gamma}'_1(t_0))^T (f \mathcal{F}_I) \bar{\gamma}'_2(t_0) = (f)(\gamma'_1(t_0) \cdot \gamma'_2(t_0))$$

Similarly, $\tilde{\gamma}'_1(t_0) = \|\bar{\gamma}'_1(t_0)\|_{\tilde{\mathcal{F}}_I} = (\sqrt{f})\|\bar{\gamma}'_1(t_0)\|_{\mathcal{F}_I}$ and $\tilde{\gamma}'_2(t_0) = \|\bar{\gamma}'_2(t_0)\|_{\tilde{\mathcal{F}}_I} = (\sqrt{f})\|\bar{\gamma}'_2(t_0)\|_{\mathcal{F}_I}$. Therefore, if $\tilde{\theta}$ is the angle between $\tilde{\gamma}'_1(t_0)$ and $\tilde{\gamma}'_2(t_0)$, we have

$$\cos \tilde{\theta} = \frac{\tilde{\gamma}'_1(t_0) \cdot \tilde{\gamma}'_2(t_0)}{\|\tilde{\gamma}'_1(t_0)\| \|\tilde{\gamma}'_2(t_0)\|} = \frac{(f)(\gamma'_1(t_0) \cdot \gamma'_2(t_0))}{((\sqrt{f})\|\gamma'_1(t_0)\|)((\sqrt{f})\|\gamma'_2(t_0)\|)} = \frac{\gamma'_1(t_0) \cdot \gamma'_2(t_0)}{\|\gamma'_1(t_0)\| \|\gamma'_2(t_0)\|} = \cos \theta$$

using the fact that f is positive so $(\sqrt{f})^2 = f$. Taking arccos, we get that $\tilde{\theta} = \theta$, proving (2). ■

Chapter 4.2: Second Fundamental Form

Recall

Let $w_1, w_2 \in \mathbb{R}^3$. The orthogonal projection of w_1 onto w_2 is $\ell = \frac{w_1 \cdot w_2}{\|w_2\|}$. In particular, if $\|w_2\| = 1$, then $\ell = w_1 \cdot w_2$.

Definition

Let S be a smooth surface in $\mathbb{R}^3$. We want to measure to what extent S “curves”, that is, to what extent S fails to be part of a (2-dimensional) plane in $\mathbb{R}^3$. Let $\sigma : U \rightarrow S \cap W$ be a smooth coordinate chart. Then for $p = \sigma(u_0, v_0) \in S \cap W$, $T_p S = \text{Span}\{\sigma_u(u_0, v_0), \sigma_v(u_0, v_0)\}S$ and $N_\sigma(u_0, v_0) = \frac{\sigma_u(u_0, v_0) \times \sigma_v(u_0, v_0)}{\|\sigma_u(u_0, v_0) \times \sigma_v(u_0, v_0)\|}$. Let $p' = \sigma(u_0 + \Delta u, v_0 + \Delta v)$ be another point in $S \cap W$. Set $\Delta \sigma = \sigma(u_0 + \Delta u, v_0 + \Delta v) - \sigma(u_0, v_0)$. Let ℓ be the orthogonal projection of $\Delta \sigma$ onto N_σ . Then $\Delta \sigma$ is orthogonal to N_σ if and only if $\ell = 0$, so $S \cap W \subseteq T_p S$ if and only if $\ell = 0$, i.e. $\ell = 0$ if and only if $S \cap W$ is contained in a plane. Note that since N_σ is a unit vector, we have that $\ell = \Delta \sigma \cdot N_\sigma$, so $\ell = (\sigma(u_0 + \Delta u, v_0 + \Delta v) - \sigma(u_0, v_0)) \cdot N_\sigma$. By Taylor’s theorem,

$$\begin{aligned} & \sigma(u_0 + \Delta u, v_0 + \Delta v) \\ &= \sigma(u_0, v_0) + \sigma_u(u_0, v_0)\Delta u + \sigma_v(u_0, v_0)\Delta v + \frac{1}{2}(\sigma_{uu}(u_0, v_0)\Delta u^2 + 2\sigma_{uv}(u_0, v_0)\Delta u\Delta v + \sigma_{vv}(u_0, v_0)\Delta v^2) + \text{remainder} \end{aligned}$$

So (omitting the arguments u_0, v_0)

$$\ell \approx (\sigma_u \Delta u + \sigma_v \Delta v + \frac{1}{2}(\sigma_{uu}\Delta u^2 + 2\sigma_{uv}\Delta u\Delta v + \sigma_{vv}\Delta v^2)) \cdot N_\sigma$$

$$\begin{aligned} & \stackrel{N_\sigma \text{ orth. to } \sigma_u, \sigma_v}{=} \frac{1}{2}(\Delta u^2(\sigma_{uu} \cdot N_\sigma) + 2\Delta u\Delta v(\sigma_{uv} \cdot N_\sigma) + \Delta v^2(\sigma_{vv} \cdot N_\sigma)) \\ &= \frac{1}{2} \begin{bmatrix} \Delta u & \Delta v \end{bmatrix} \begin{bmatrix} \sigma_{uu} \cdot N_\sigma & \sigma_{uv} \cdot N_\sigma \\ \sigma_{vu} \cdot N_\sigma & \sigma_{vv} \cdot N_\sigma \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta v \end{bmatrix} \end{aligned}$$

The matrix $\begin{bmatrix} \sigma_{uu} \cdot N_\sigma & \sigma_{uv} \cdot N_\sigma \\ \sigma_{vu} \cdot N_\sigma & \sigma_{vv} \cdot N_\sigma \end{bmatrix}$ is denoted $\mathcal{F}_{II}$, and is called the second fundamental form of σ .

Example

Let σ be a global chart of a plane given by $\sigma(u, v) = p + uw_1 + vw_2$, so $\sigma_u = w_1, \sigma_v = w_2$, so

$\sigma_{uu} = \sigma_{uv} = \sigma_{vu} = \sigma_{vv} = 0$, so $\mathcal{F}_{II} = 0$.

Example

Let σ be the chart we have seen of the cylinder before, $\sigma(u, v) = (\cos u, \sin u, v)$, $(u, v) \in (0, 2\pi) \times \mathbb{R}$. Then $\sigma_u = (-\sin u, \cos u, 0)$, $\sigma_v = (0, 0, 1)$, $\sigma_{uu} = (-\cos u, -\sin u, 0)$, and $\sigma_{uv} = \sigma_{vu} = \sigma_{vv} = 0$. Thus $\sigma_{uu} \cdot N_\sigma = (-\cos u, -\sin u, 0) \cdot (\cos u, \sin u, 0) = -1$, so $\mathcal{F}_{II} = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$.

Definition

The shape operator (or Weingarten matrix) is defined to be $\mathcal{W} := \mathcal{F}_I^{-1} \mathcal{F}_{II}$, and the Gaussian curvature is $K := \det \mathcal{W}$, and the mean curvature is $H := \frac{1}{2} \operatorname{tr} \mathcal{W}$. Let $p = \sigma(u_0, v_0)$. Then:

- (1) p is elliptic if $K > 0$.
- (2) p is hyperbolic if $K < 0$.
- (3) p is parabolic if $K = 0, H \neq 0$.
- (4) p is planar if $K = H = 0$.

Example

On a plane parameterized in the usual way, $\mathcal{F}_I = I_{2 \times 2}$ and $\mathcal{F}_{II} = 0$, so $\mathcal{W} = 0$, and thus $K = H = 0$ so every point on the plane is planar.

Example

On a cylinder as parameterized above, $\mathcal{F}_I = I_{2 \times 2}$, $\mathcal{F}_{II} = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$, so $\mathcal{W} = \mathcal{F}_{II}$, so $K = 0$ and $H = -\frac{1}{2}$, so every point on the cylinder is parabolic.

Example

Let $S = \{(x, y, z) : z = x^2\}$ be a parabolic cylinder. Then $\sigma(u, v) = \sigma(u, v, u^2)$ is a chart, where $\mathcal{W} = \begin{bmatrix} \frac{2}{(1+4u^2)^{3/2}} & 0 \\ 0 & 0 \end{bmatrix}$, where we see $K = 0$ and $H = \frac{1}{(1+4u^2)^{3/2}} \neq 0$, so every point on S is parabolic.

Example

For $a > 0$, let $x^2 + y^2 + z^2 = a^2$ be the sphere S^2 . Then a chart is $\sigma(\theta, \phi) = (a \sin \phi \cos \theta, a \sin \phi \sin \theta, a \cos \phi)$ for $(\theta, \phi) \in (0, 2\pi) \times (0, \pi)$, where $\mathcal{W} = \begin{bmatrix} 1/a & 0 \\ 0 & 1/a \end{bmatrix}$ so $K = 1/a^2$, and $H = 1/a$ which are both constant (and we see the sphere is elliptic at every point).

Example

Let $S = \{(x, y, z) \in \mathbb{R}^3 : z = x^2 - y^2\}$ be the saddle surface, with a global chart $\sigma(u, v) = (u, v, u^2 - v^2)$. Then $H = \det \mathcal{W} = \det(\mathcal{F}_{II}) / \det(\mathcal{F}_I)$, where $\mathcal{F}_I = \begin{bmatrix} 1+4u^2 & -4uv \\ -4uv & 1+4v^2 \end{bmatrix}$, $\mathcal{F}_{II} = \begin{bmatrix} \frac{1}{\sqrt{1+4u^2+4v^2}} & 2 \\ 0 & -2 \end{bmatrix}$, so $K = \frac{-1}{(1+4u^2+4v^2)^2} < 0$, and so every point on the saddle surface is hyperbolic.

Example

Suppose $f : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ is a smooth scalar function, where S is its graph. Then a global chart for S is $\sigma(u, v) = (u, v, f(u, v))$, with $\mathcal{F}_I = \begin{bmatrix} 1+f_u^2 & f_u f_v \\ f_u f_v & 1+f_v^2 \end{bmatrix}$, with $N_\sigma = \frac{1}{\sqrt{1+f_u^2+f_v^2}}(-f_u, -f_v, 1)$, so $\mathcal{F}_{II} = \frac{1}{\sqrt{1+f_u^2+f_v^2}} \begin{bmatrix} f_{uu} & f_{uv} \\ f_{vu} & f_{vv} \end{bmatrix} = \frac{1}{\sqrt{1+f_u^2+f_v^2}} H(f)$, where $H(f) = \begin{bmatrix} f_{uu} & f_{uv} \\ f_{vu} & f_{vv} \end{bmatrix}$ is the Hessian matrix of f , and so $\mathcal{W} = \frac{1}{(1+f_u^2+f_v^2)^{3/2}} \begin{bmatrix} 1+f_v^2 & -f_u f_v \\ -f_v f_u & 1+f_u^2 \end{bmatrix} H(f)$. Recall a point $(u_0, v_0) \in U$ is a critical point if $f_u(u_0, v_0) = f_v(u_0, v_0) = 0$. In this case, $\mathcal{F}_I(u_0, v_0) = I_{2 \times 2}$ and $\mathcal{F}_{II}(u_0, v_0) = H(f)(u_0, v_0)$, so $\mathcal{W}(u_0, v_0) = H(f)(u_0, v_0)$. Then by the second derivative test we know if $\det \mathcal{W}(u_0, v_0) = K(u_0, v_0) > 0$, then we have a local maximum or minimum, which looks locally like the pole of an ellipse, agreeing with

our definition of an elliptic point. If $\det \mathcal{W}(u_0, v_0) = K(u_0, v_0) < 0$, then we have a saddle point by the second derivative test, which is hyperbolic, so our definition agrees here as well.

Why do we use $\mathcal{W}$ instead of just $\mathcal{F}_{II}$ to study curvature?

$\mathcal{F}_{II}$ is not invariant under change of coordinate chart. Suppose that $\tilde{\sigma} : \tilde{U} \rightarrow S \cap \tilde{W}$ is another coordinate chart with $\tilde{\sigma} = \sigma \circ \Phi$ for Φ a diffeomorphism. We have seen that $\tilde{\mathcal{F}}_I = D\Phi^T \mathcal{F}_I D\Phi$. A similar but messier computation shows, $\tilde{\mathcal{F}}_{II} = \pm D\Phi^T \mathcal{F}_{II} D\Phi$, so in general, $\det \mathcal{F}_{II} \neq \det \tilde{\mathcal{F}}_{II}$ and $\text{tr } \mathcal{F}_{II} \neq \text{tr } \tilde{\mathcal{F}}_{II}$. But then this shows

$$\begin{aligned} \tilde{\mathcal{W}} &= \tilde{\mathcal{F}}_I^{-1} \tilde{\mathcal{F}}_{II} = \pm (D\Phi^T \mathcal{F}_I D\Phi)^{-1} (D\Phi^T \mathcal{F}_{II} D\Phi) = \pm D\Phi^{-1} \mathcal{F}_I^{-1} D\Phi^{T^{-1}} D\Phi^T \mathcal{F}_{II} D\Phi \\ &= \pm D\Phi^{-1} \mathcal{F}_I^{-1} \mathcal{F}_{II} D\Phi = \pm D\Phi^{-1} \mathcal{W} D\Phi \end{aligned}$$

so $\tilde{\mathcal{W}}$ is conjugate to $\mathcal{W}$ up to a sign (depending on the normal chosen), and thus shares the same eigenvalues and trace up to a sign, and does share the same determinant (because the relevant matrices are 2×2), so K is invariant under the choice of chart, and H is invariant up to sign.

Note that although $\mathcal{F}_I$ and $\mathcal{F}_{II}$ are symmetric matrices, $\mathcal{W} = \mathcal{F}_I^{-1} \mathcal{F}_{II}$ may not be symmetric. Nonetheless, there exists a matrix $C \in \text{GL}(2, \mathbb{R})$ and a symmetric matrix B such that $B = C\mathcal{W}C^{-1}$ (to be proved later), which tells us $\mathcal{W}$ is diagonalizable and has real eigenvalues by the spectral theorem. We thus obtain:

Proposition/Definition

$\mathcal{W}$ is always diagonalizable with real eigenvalues κ_1, κ_2 , which are called the principal curvatures of S , and are independent up to sign of the parameterization of S , where $K = \kappa_1 \kappa_2$ and $H = \frac{\kappa_1 + \kappa_2}{2}$. So if $p_0 = \sigma(u_0, v_0)$, then p_0 is elliptic if κ_1 and κ_2 have the same sign, it is hyperbolic if they have opposite signs, it is parabolic if exactly one of them is zero, and it is planar if both are zero. ■

Let us now prove that $\mathcal{W}$ is conjugate to a symmetric matrix:

Definition

Given a smooth surface S with smooth chart $\sigma : U \rightarrow S \cap W$, the Gauss map $g_\sigma : U \subseteq \mathbb{R}^2 \rightarrow S^2$ is the map $(u, v) \mapsto N_\sigma(u, v)$, where since N_σ has norm 1 everywhere, this map is well-defined, and is smooth. Moreover, $Dg_\sigma = [\frac{\partial}{\partial u} N_\sigma | \frac{\partial}{\partial v} N_\sigma]$, and note that since N_σ has norm 1 everywhere, it follows that its partial derivatives are orthogonal to it, so $\frac{\partial}{\partial u} N_\sigma, \frac{\partial}{\partial v} N_\sigma \in T_p S$.

Proposition (Theorema Egregium)

There exists a symmetric matrix B and a matrix $C \in \text{GL}(2, \mathbb{R})$ such that $B = C\mathcal{W}C^{-1}$. In particular, $\mathcal{W}$ is diagonalizable with real eigenvalues by the spectral theorem.

Proof:

We want to show $\mathcal{W}$ is the matrix representation of a self-adjoint operator. Following the above construction of the partial derivatives of N_σ , we define the Weingarten map

$$\mathcal{L}_\sigma : T_p S \rightarrow T_p S$$

defined by

$$(s\sigma_u + t\sigma_v) \mapsto -\left(s\frac{\partial}{\partial u} N_\sigma + t\frac{\partial}{\partial v} N_\sigma\right) = -Dg_\sigma \begin{bmatrix} s \\ t \end{bmatrix}$$

which is linear by our above matrix representation. We claim that this is self-adjoint and has matrix representation $\mathcal{W}$ with respect to the basis $\mathcal{B} := \{\sigma_u, \sigma_v\}$. Let us first prove $[\mathcal{L}_\sigma]_{\mathcal{B}} = \mathcal{W}$. Since $\frac{\partial}{\partial u} N_\sigma, \frac{\partial}{\partial v} N_\sigma \in T_p S$, we have $\frac{\partial}{\partial u} N_\sigma = a\sigma_u + b\sigma_v$, $\frac{\partial}{\partial v} N_\sigma = c\sigma_u + d\sigma_v$, so $\mathcal{L}_\sigma(\sigma_u) = -\frac{\partial}{\partial u} N_\sigma = -(a\sigma_u + b\sigma_v)$, and similarly $\mathcal{L}_\sigma(\sigma_v) = -(c\sigma_u + d\sigma_v)$, so all-in-all $[\mathcal{L}_\sigma]_{\mathcal{B}} = -\begin{bmatrix} a & c \\ b & d \end{bmatrix}$. Moreover, $N_\sigma \cdot \sigma_u = N_\sigma \cdot \sigma_v = 0$, so

differentiating, we get

$$0 = \frac{\partial}{\partial u}(N_\sigma \cdot \sigma_u) = \left(\frac{\partial}{\partial u} N_\sigma\right) \cdot \sigma_u + N_\sigma \cdot \sigma_{uu} = (a\sigma_u + b\sigma_v) \cdot \sigma_u + N_\sigma \cdot \sigma_{uu} = a(\sigma_u \cdot \sigma_u) + b(\sigma_v \cdot \sigma_u) + N_\sigma \cdot \sigma_{uu}$$

Since $\sigma_u \cdot \sigma_u$ and $\sigma_v \cdot \sigma_u$ are the matrix entries of $\mathcal{F}_I$, and $N_\sigma \cdot \sigma_{uu}$ is an entry of $\mathcal{F}_{II}$. We get a similar equation from $0 = \frac{\partial}{\partial u}(N_\sigma \cdot \sigma_v)$. Combining these, we get:

$$\mathcal{F}_I \begin{bmatrix} a \\ b \end{bmatrix} + \mathcal{F}_{II} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0$$

Similarly differentiating $N_\sigma \cdot \sigma_u$ and $N_\sigma \cdot \sigma_v$ with respect to v give

$$\mathcal{F}_I \begin{bmatrix} c \\ d \end{bmatrix} + \mathcal{F}_{II} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

Combining these tells us

$$\mathcal{F}_I \begin{bmatrix} a & c \\ b & d \end{bmatrix} + \mathcal{F}_{II} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

which tells us $\mathcal{F}_{II} = -\mathcal{F}_I \begin{bmatrix} a & c \\ b & d \end{bmatrix}$, so $[\mathcal{L}]_{\mathcal{B}} = -\begin{bmatrix} a & c \\ b & d \end{bmatrix} = \mathcal{F}_I^{-1} \mathcal{F}_{II} = \mathcal{W}$. To show $\mathcal{L}_\sigma$ is self-adjoint, let $X, Y \in T_p S$:

$$\begin{aligned} \mathcal{L}_\sigma(X) \cdot Y &= [\mathcal{L}_\sigma(X)]_{\mathcal{B}}^T \mathcal{F}_I[Y]_{\mathcal{B}} = ([\mathcal{L}_\sigma]_{\mathcal{B}}[X]_{\mathcal{B}})^T \mathcal{F}_I[Y]_{\mathcal{B}} = [X]_{\mathcal{B}}^T [\mathcal{L}_\sigma]_{\mathcal{B}} \mathcal{F}_I[Y]_{\mathcal{B}} \\ &= [X]_{\mathcal{B}}^T \mathcal{W}^T \mathcal{F}_I[Y]_{\mathcal{B}} = [X]_{\mathcal{B}}^T \mathcal{F}_{II}^T \mathcal{F}_I^{-1^T} \mathcal{F}_I[Y]_{\mathcal{B}} \end{aligned}$$

Since $\mathcal{F}_I$ is symmetric, $\mathcal{F}_I^{-1^T} = \mathcal{F}_I^{-1}$, so this is equal to (since $\mathcal{F}_{II}$ is symmetric)

$$\begin{aligned} &= [X]_{\mathcal{B}}^T \mathcal{F}_{II} \mathcal{F}_I[Y]_{\mathcal{B}} = [X]_{\mathcal{B}}^T \mathcal{F}_I \mathcal{F}_I^{-1} \mathcal{F}_{II} \mathcal{F}_I[Y]_{\mathcal{B}} \\ &= [X]_{\mathcal{B}}^T \mathcal{F}_I \mathcal{W} \mathcal{F}_I[Y]_{\mathcal{B}} = [X]_{\mathcal{B}}^T \mathcal{F}_I [L_\sigma(Y)]_{\mathcal{B}} = X \cdot L_\sigma(Y) \end{aligned}$$

so $\mathcal{L}_\sigma$ is self-adjoint. ■

Corollary

For each $p \in S$, there exists an orthonormal basis of eigenvectors $\{X_1, X_2\}$ of $\mathcal{L}_\sigma$ corresponding to eigenvalues κ_1, κ_2 for $T_p S$.

Proof:

This is just the spectral theorem, but more explicitly, since L_σ is self-adjoint, its eigenspaces corresponding to different eigenvalues are orthogonal, so normalizing them we get such a basis. ■

Chapter 4.6(?): Geodesic and normal curvatures

Let $S \subseteq \mathbb{R}^3$ be a smooth surface, $\sigma : U \rightarrow S \cap W$ be a chart, and $p \in S \cap W$ a point in the image of the chart. Suppose that $p = \sigma(u_0, v_0)$. Let $\gamma : (\alpha, \beta) \rightarrow S \cap W$ be a smooth unit-speed curve in $S \cap W$ such that $\gamma(t_0) = p$ for some $t_0 \in (\alpha, \beta)$. We claim $\gamma''(t_0) \in \text{Span}_{\mathbb{R}}\{N_\sigma(u_0, v_0), N_\sigma(u_0, v_0) \times \gamma'(t_0)\}$. Indeed, if it is zero then this holds trivially, so suppose $\gamma''(t_0) \neq 0$. Then since γ is unit speed, and $\gamma''(t_0) \neq 0$, then $\gamma''(t_0)$ is orthogonal to $\gamma'(t_0)$. But $\gamma'(t_0) \in T_p S$, which is orthogonal to $N_\sigma(u_0, v_0)$ and $N_\sigma(u_0, v_0) \times \gamma'(t_0)$. Thus, for dimension reasons, this forces $\gamma''(t_0)$ to be in the span of $N_\sigma(u_0, v_0)$ and $N_\sigma(u_0, v_0) \times \gamma'(t_0)$ (note the cross product is nonzero since $\gamma'(t_0) \neq 0$).

Definition

In light of the above, for a curve γ with the above setup, we can write $\gamma''(t_0) = \kappa_n N_\sigma(u_0, v_0) +$

$\kappa_g(N_\sigma(u_0, v_0) \times \gamma'(t_0))$ for some real numbers $\kappa_n, \kappa_g \in \mathbb{R}$, which we call the normal and geodesic curvatures of γ respectively. We call γ a geodesic if $\kappa_g \equiv 0$.

Note that since N_σ and $\gamma'(t_0)$ are unit length, so is $N_\sigma \times \gamma'(t_0)$, so $\kappa_n = \gamma''(t_0) \cdot N_\sigma$ and $\kappa_g = \gamma''(t_0) \cdot (N_\sigma \times \gamma'(t_0))$. Then the usual curvature $\kappa = \|\gamma''(t_0)\|$ satisfies $\kappa^2 = \kappa_n^2 + \kappa_g^2$ by the Pythagorean theorem. The vector $\kappa_n N_\sigma$ is called the normal component of γ .

Moreover, note that an equivalent definition of geodesics is that γ'' is parallel to N_σ (or 0), and thus must be perpendicular to $T_p S = \text{Span}_{\mathbb{R}}\{\sigma_u, \sigma_v\}$, so γ is a geodesic if and only if $\gamma''(t_0) \cdot \sigma_u(u_0, v_0) = 0 = \gamma''(t_0) \cdot \sigma_v(u_0, v_0)$ for all t_0 . These are called the geodesic equations. If $\gamma(t) = \sigma(u(t), v(t))$ for smooth functions, $u, v : (\alpha, \beta) \rightarrow \mathbb{R}$ is a curve on S , we can rewrite these as

$$\begin{aligned} \frac{d}{dt}(Eu' + Fv') &= \frac{1}{2} \begin{bmatrix} u' & v' \end{bmatrix} \begin{bmatrix} E_u & F_u \\ F_u & G_u \end{bmatrix} \begin{bmatrix} u' \\ v' \end{bmatrix} \\ \frac{d}{dt}(Fu' + Gv') &= \frac{1}{2} \begin{bmatrix} u' & v' \end{bmatrix} \begin{bmatrix} E_v & F_v \\ F_v & G_v \end{bmatrix} \begin{bmatrix} u' \\ v' \end{bmatrix} \end{aligned}$$

where $\mathcal{F}_I = \begin{bmatrix} E & F \\ F & G \end{bmatrix}$. This is an ordinary differential equation in terms of $\mathcal{F}_I$, and so always has a solution, so in particular, geodesics for S always exist! Because they are in terms of $\mathcal{F}_I$ only, they are preserved by isometries.

Proposition

The normal curvature κ_n of a curve $\gamma : (\alpha, \beta) \rightarrow S$ at $t \in (\alpha, \beta)$ is given by $\mathcal{L}_\sigma(\gamma'(t)) \cdot \gamma'(t)$.

Proof:

Recall $\bar{\gamma} := \sigma^{-1} \circ \gamma$, so $N_\sigma(\bar{\gamma}(t)) \cdot \gamma'(t) = 0$, giving that

$$0 = \frac{d}{dt}(N_\sigma(\bar{\gamma}(t)) \cdot \gamma'(t)) = \frac{d}{dt}(N_\sigma(\bar{\gamma}(t))) \cdot \gamma'(t) + N_\sigma(\bar{\gamma}(t)) \cdot \gamma''(t)$$

Now, by the chain rule,

$$\frac{d}{dt}(N_\sigma(\bar{\gamma}(t))) = DN_\sigma(\bar{\gamma}(t))\bar{\gamma}'(t) = -\mathcal{L}_\sigma(\gamma'(t))$$

Moreover, by definition $N_\sigma(\bar{\gamma}(t)) \cdot \gamma''(t) = \kappa_n$ so altogether this shows

$$-\mathcal{L}_\sigma(\gamma'(t)) \cdot \gamma'(t) + \kappa_n = 0$$

as desired. ■

Corollary

Let $\{X_1, X_2\}$ be an orthonormal basis for $T_p S$ such that $\mathcal{L}(X_i) = \kappa_i X_i$ (by the spectral theorem/above corollary). Then

- (1) $\gamma'(t_0) = \cos(\theta_0)X_1 + \sin(\theta_0)X_2$ where θ_0 is the angle between $\gamma'(t_0)$ and X_1 .
- (2) $\kappa_n = \cos^2 \theta_0 \kappa_1 + \sin^2 \theta_0 \kappa_2$. In particular, $\kappa_n \leq \max\{\kappa_1, \kappa_2\}$ and $\kappa_n \geq \min\{\kappa_1, \kappa_2\}$.

Proof:

(1) follows because X_1, X_2 is an orthonormal basis of $T_p S$. For (2), apply the above proposition to get $\kappa_n = \mathcal{L}_\sigma(\gamma'(t_0)) \cdot \gamma'(t_0)$, and so since X_1 and X_2 are eigenvectors, we have

$$\begin{aligned} \mathcal{L}_\sigma(\gamma'(t_0)) &= \mathcal{L}_\sigma(\cos(\theta_0)X_1 + \sin(\theta_0)X_2) = \cos(\theta_0)\mathcal{L}_\sigma(X_1) + \sin(\theta_0)\mathcal{L}_\sigma(X_2) \\ &= \cos(\theta_0)\kappa_1 X_1 + \sin(\theta_0)\kappa_2 X_2 \end{aligned}$$

so by (1) and the fact X_1, X_2 are orthonormal,

$$\kappa_n = (\kappa_1 \cos(\theta_0)X_1 + \kappa_2 \sin(\theta_0)X_2) \cdot (\cos(\theta_0)X_1 + \sin(\theta_0)X_2) = \cos^2(\theta_0)\kappa_1 + \sin^2(\theta_0)\kappa_2 \quad \blacksquare$$

Example

A line $\gamma(t) = p_0 + tw$, $t \in \mathbb{R}$ in a plane is always a geodesic because its second derivative vanishes.

Example

Let S^2 be the sphere $x^2 + y^2 + z^2 = a^2$, $a > 0$. Then $\sigma(\theta, \phi) = (a \cos \theta \sin(\phi/a), a \sin \theta \sin(\phi/a), a \cos(\phi/a))$ for $(\theta, \phi) \in (0, 2\pi) \times (a\pi)$. Then $N_\sigma(\theta, \phi) = -(\cos \theta \sin(\phi/a), \sin \theta \sin(\phi/a), \cos(\phi/a))$. Consider a great circle C of radius a , which is the curve $\gamma(\phi) = \sigma(\theta_0, \phi)$ for fixed θ_0 . This is the intersection of the sphere with the vertical plane $\theta = \theta_0$. Then $\|\gamma'(\phi)\| = 1$ for all ϕ , and $\gamma''(\phi) = -\frac{1}{a}(\cos \theta_0 \sin(\phi/a), \sin \theta_0 \sin(\phi/a), \cos(\phi/a))$, which one sees is equal to $\frac{1}{a}N_\sigma(\theta_0, \phi)$ and thus parallel, so it is a geodesic for the sphere.

Example

Consider a plane with global chart $\sigma(u, v) = p_0 + uw_1 + vw_2$ for all $(u, v) \in \mathbb{R}^2$, for orthonormal vectors $w_1, w_2 \in \mathbb{R}^3$. Then $\mathcal{F}_I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, so the geodesic equations become

$$\frac{d}{dt}(u') = 0 = \frac{d}{dt}(v')$$

which implies u' and v' are constant, which integrating implies $u(t) = at + b$ and $v(t) = ct + d$ for $a, b, c, d \in \mathbb{R}$ so writing $\gamma(t) = \sigma(u(t), v(t))$, we have $\gamma(t) = p_0 + (at + b)w_1 + (ct + d)w_2$ for all $t \in \mathbb{R}$, which is a line. Thus any geodesic on a plane is a line, so by the example above, the geodesics on a plane is exactly the set of straight lines.

Example

For the cylinder $x^2 + y^2 = 1$, with chart $\sigma(u, v) = (\cos u, \sin u, v)$ where $(u, v) \in (0, 2\pi) \times \mathbb{R}$, we have $\mathcal{W} = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$, with eigenvalues $\kappa_1 = -1, \kappa_2 = 0$. Then $X_1 = (\sin u, \cos u)$, $X_2 = \sigma_v = (0, 0, 1)$ is an orthonormal basis of $T_p S$ made of eigenvectors of $\mathcal{W}$. Then X_1 is the velocity vector of a horizontal circular path on the cylinder, with constant curvature $\kappa = 1 = |\kappa_1|$ and X_2 is the velocity of a vertical line on the cylinder, with constant curvature $\kappa = 0 = |\kappa_2|$.

Example

We know the cylinder is isometric to an infinite strip in $\mathbb{R}^2$ and geodesics are preserved by isometries. Thus every geodesic on the cylinder corresponds to the image of a straight line under such an isometry, which turn out to be circles, vertical lines, and helices. In particular, write $S = \{(x, y, 0) : 0 < x < 2\pi, z = 0\}$, and $S' = \{x^2 + y^2 = 1\} \setminus \{(1, 0, z) : z \in \mathbb{R}\}$. Then $F : S \rightarrow S'$ given by $(x, y, 0) \mapsto (\cos x, \sin x, y)$ is an isometry. But the geodesics on planes are lines, given a parameterization $\gamma(t) = (at + b, ct + d, 0)$ for some $a, b, c, d \in \mathbb{R}$, which corresponds to a geodesic $\tilde{\gamma}(t) = F \circ \gamma(t) = F(at + b, ct + d, 0)$, so $\tilde{\gamma}(t) = (\cos(at + b), \sin(at + b), ct + d)$ for all $t \in \mathbb{R}$ is exactly all of the geodesics on the cylinder. If $a = 0$, then this is a vertical line through the point $(\cos b, \sin b, d)$, specifically $\tilde{\gamma}(t) = (\cos b, \sin b, ct + d)$, if $c = 0$, then this is a circle through $z = d$, namely $\tilde{\gamma}(t) = (\cos(at + b), \sin(at + b), d)$, and if $a \neq 0, c \neq 0$, then this is a helix.

Geodesics as shortest paths

Let $p, q \in S$. We want to find the curve $\gamma(-2, 2) \rightarrow S$ such that $p = \gamma(0), q = \gamma(1)$, and γ has the smallest arc length. Since γ is on S , $\gamma(t) = \sigma(u(t), v(t))$ for smooth functions $u, v : (-2, 2) \rightarrow \mathbb{R}$. Let $\delta > 0$. Then, for all $\varepsilon \in (-\delta, \delta)$, set $\gamma_\varepsilon(t) := \sigma(u(t) + \varepsilon\alpha(t), v(t) + \varepsilon\beta(t))$, where $\alpha, \beta : (-2, 2) \rightarrow \mathbb{R}$ are smooth functions such that $\alpha(0) = \beta(0) = \alpha(1) = \beta(1) = 0$. Then, $\gamma_\varepsilon(0) = \gamma(0) = p$, and $\gamma_\varepsilon(1) = \gamma(1) = q$. This gives us a 1-parameter family of curves on S with initial point p and terminal point q such that $\gamma_0 = \gamma$. Define the arclength of γ_ε from p to q as $L(\varepsilon) := \int_0^1 \|\gamma'_\varepsilon(t)\| dt$. Then, γ minimizes arc length only if $\varepsilon = 0$ is a critical point of $L(\varepsilon)$, i.e. $L'(0) = 0$ (for every initial choices of α, β). Doing some fun

calculus (ALEX WRITE THIS UP LATER), we have

$$L'(0) = - \int_0^1 (\alpha(t)(\gamma''(t) \cdot \sigma_u) + \beta(t)(\gamma''(t) \cdot \sigma_v)) dt$$

This is equal to 0 for every α, β if and only if $\gamma''(t) \cdot \sigma_u = \gamma''(t) \cdot \sigma_v = 0$, i.e. γ is a geodesic. Therefore, any curve that minimizes distance is a geodesic. The converse is not true, for example, great circles are geodesics on the sphere, but if one follows the great circle the long way between two points, this is obviously not the shortest path between two points.

Here's another reason we care about geodesics. If we draw a "geodesic triangle" (three points with the three geodesics joining them between them as edges), with angles α, β, γ , then $\alpha + \beta + \gamma = \pi + \int_T K dA$, where K is the Gaussian curvature (and we will define this integral later) and we integrate over the triangle T . In particular, if $K \equiv 0$, for example if T is part of a plane, then $\alpha + \beta + \gamma = \pi$, and the sides of the geodesic triangle are line segments, so this is the fact about the sum of the angles of a triangle from high school. If $K > 0$ everywhere, then $\int_T K dA > 0$, implying $\alpha + \beta + \gamma > \pi$, for example if S is a sphere of radius a , then $K \equiv \frac{1}{a^2} > 0$, so on the sphere, $\alpha + \beta + \gamma > \pi$. One can use this fact about geodesic triangles to prove the following:

Theorem (Gauss-Bonnet)

Let S be a compact smooth orientable surface. Then $\int_S K dA = 4\pi(1 - g)$, where g is the genus of S ("the number of holes").

This is a remarkable fact! The genus is a topological invariant of S , but K is a differential geometric object that changes as we deform S or give it a different chart!

But what do the integrals actually mean for a submanifold?

Chapter 5: Integration on Submanifolds

Let M be a smooth k -submanifold of $\mathbb{R}^n$, with $\sigma : U \rightarrow M \cap W$ a smooth chart. Then, for all $p \in M \cap W$ with $u = \sigma^{-1}(p) \in \mathbb{R}^k$, we have $T_p M = \text{Span}_{\mathbb{R}}\{\frac{\partial \sigma}{\partial u_1}(u), \dots, \frac{\partial \sigma}{\partial u_k}(u)\}$. Then, define the first fundamental form $\mathcal{F}_I = D\sigma^T D\sigma = [\frac{\partial \sigma}{\partial u_i} \cdot \frac{\partial \sigma}{\partial u_j}]_{i,j}$. Then, as a positive-definite symmetric $k \times k$ matrix, $\det \mathcal{F}_I$ is positive, so we may consider $\sqrt{\det \mathcal{F}_I}$. How can we interpret this quantity?

Example

Suppose that $k = 1$, so that M is a regular curve $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^n$. Then $\mathcal{F}_I = \gamma'(t) \cdot \gamma'(t) = \|\gamma'(t)\|^2$, so $\sqrt{\det \mathcal{F}_I} = \|\gamma'(t)\|$. If $k = 2$, then $\sqrt{\det \mathcal{F}_I} = \|\sigma_u \times \sigma_v\|$, which is the area of the parallelogram spanned by σ_u and σ_v , and in general $\sqrt{\det \mathcal{F}_I}$ is the volume of the parallelepiped spanned by $\sigma_{u_1}, \dots, \sigma_{u_k}$.

Definition

Let $f : M \rightarrow \mathbb{R}$ be a continuous function. We define, when $M \cap W$ is bounded,

$$\int_{M \cap W} f dVol := \int_{U \subseteq \mathbb{R}^k} f(\sigma(u)) \sqrt{\det \mathcal{F}_I} du_1 \dots du_k$$

Example

If $k = 1$, then M is a regular curve $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^n$ and this integral is a line integral, $\int_{\gamma} f ds := \int_M f dVol := \int_{\alpha}^{\beta} f(\gamma(t)) \|\gamma'(t)\| dt$, where if $f \equiv 1$, this is the arc length of γ . If $k = 2$, then this is the integral $\int_S f dA := \int_U f(\sigma(u, v)) \|\sigma_u \times \sigma_v\| du dv$, and if $k = n$, then this is the usual multiple integral $\int_M f dVol = \int_M f(u_1, \dots, u_k) du_1 \dots du_k$.

Integrating a Vector Field

Definition

A vector field on M is a function $F : M \rightarrow \mathbb{R}^n$, usually smooth.

Example

If we take $M = \mathbb{R}^3$, all smooth functions $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ are smooth vector fields, for example $(x, y, z) \mapsto (xy, z + x, e^x y)$.

Example

Let $M \subseteq \mathbb{R}^3$ be a smooth surface and $\sigma : U \rightarrow M \cap W$ be a smooth chart. Define $N : M \cap W \rightarrow \mathbb{R}^3$ by $N(p) = N_\sigma(\sigma^{-1}(p))$ for all $p \in M \cap W$. Then this is a vector field on $M \cap W$, and it is normal to $T_p M$ for all $p \in M \cap W$, so it is called a normal vector field.

Example

Let M be a 1-submanifold of $\mathbb{R}^n$, so that M is locally given by a regular curve $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^n$. Then $F := \gamma' \circ \gamma^{-1}$ is a smooth vector field on M that is tangent to M at every point.

Can we integrate a vector field on M ? Let $F : M \rightarrow \mathbb{R}^n$ be a smooth vector field. If there exists another smooth vector field $G : M \rightarrow \mathbb{R}^n$, we can consider $f(p) := F(p) \cdot G(p)$ for all $p \in M$, which is a smooth scalar function on M , which we can integrate. So, if there is a “canonical” choice for G , we can define the integral of F to be that.

Definition

Up to sign, there is a canonical choice if $\dim M = 1$ or $\dim M = n - 1$.

(1) If $\dim M = 1$ so that M is a regular curve γ locally, pick $G := T$ to be the unit tangent vector field, and set $\int_\gamma F \cdot ds = \int_\alpha^\beta F(\gamma(t)) \cdot T(t) dt$, which is uniquely defined by the curve up to a sign, the orientation of T .

(2) If $\dim M = n - 1$, and M is orientable (see the assignments), then there exists a nowhere vanishing unit normal vector field $N : M \rightarrow \mathbb{R}^n$. Then we can define $\int_M F \cdot dVol := \int_M (F \cdot N) dVol$.